

INDUSTRIAL INTERNSHIP REPORT

OPEN CLINICAL RECORD

Design and Implementation of a Focused
Outpatient Electronic Medical Record MVP

Software Engineering · Industrial Practice
HiLCoE School of Computer Science and Technology

Host Organisation

Infinity Advanced Technology Solutions PLC

Addis Ababa, Ethiopia

Internship period

01 September 2026 – 30 September 2026

Repository: github.com/elan-thinks/open-clinical-record

Academic report · Software engineering case study · Technical documentation

DECLARATION

I hereby declare that this Industrial Internship Report entitled **Open Clinical Record (OCR): Design and Implementation of a Focused Outpatient Electronic Medical Record MVP** is my original work, prepared during the industrial internship period from 01 September 2026 to 30 September 2026 at Infinity Advanced Technology Solutions PLC, Addis Ababa, Ethiopia.

The software system described herein was developed as part of the internship project under industrial mentoring. Where external literature, open-source references, or organisational guidance informed the work, sources are cited. No part of this report has been submitted previously for any academic award except as required by the host institution and HiLCoE School of Computer Science and Technology.

All clinical data used for demonstration and testing is synthetic. No real patient records were accessed, stored, or disclosed during the internship.

Student Name · Software Engineering · HiLCoE

Date: _____

APPROVAL / CERTIFICATION

This report is submitted in partial fulfilment of the requirements for the Industrial Practice / Internship component of the Software Engineering programme at HiLCoE School of Computer Science and Technology.

Industrial Supervisor / Mentor

Name: _____

Organisation: Infinity Advanced Technology Solutions PLC

Signature: _____ Date: _____

Academic Supervisor

Name: _____

Institution: HiLCoE School of Computer Science and Technology

Signature: _____ Date: _____

Host Organisation Endorsement

This certifies that the student completed an industrial internship attachment with Infinity Advanced Technology Solutions PLC for the period stated, and that the project work described is consistent with activities undertaken under organisational supervision. [EVIDENCE TO BE INSERTED — formal letterhead if required]

ACKNOWLEDGEMENTS

I thank Infinity Advanced Technology Solutions PLC for hosting this internship and for providing a professional environment in which a focused Electronic Medical Record (EMR) MVP could be researched, designed, implemented, and tested.

I am grateful to the industrial mentor(s) and engineering staff who reviewed architecture decisions, challenged incomplete assumptions about clinical workflow, and insisted on evidence over presentation. Their feedback shaped the service-layer refactoring, longitudinal visit model, and testing posture described in this report.

I acknowledge HiLCoE School of Computer Science and Technology for the academic framework of the industrial practice programme, and for emphasising that internship outputs must be both technically sound and honestly scoped.

Finally, I recognise the broader open EMR community (including design ideas observed in systems such as OpenMRS-style longitudinal visit models and published EMR workflow literature). External systems informed analysis; they are not claimed as features of Open Clinical Record.

ABSTRACT

This report documents a one-month software engineering internship (01–30 September 2026) at Infinity Advanced Technology Solutions PLC, Addis Ababa, during which the student designed and implemented **Open Clinical Record (OCR)**—a focused outpatient Electronic Medical Record (EMR) minimum viable product (MVP).

OCR addresses three core clinical-operational modules: **Patient Management**, **Patient Chart**, and **Appointment Management**. The system is implemented as a React 19 / TypeScript / Vite single-page application consuming an ASP.NET Core 8 REST API, with PostgreSQL persistence through Entity Framework Core 8 and Npgsql. Security is based on ASP.NET Core Identity, JWT Bearer authentication, and role-based authorization across four application roles: Receptionist, Nurse, Doctor, and Admin.

A central engineering contribution is the explicit separation of **Patient**, **Appointment**, and **Clinical Visit** concepts. Check-in creates a Draft clinical visit linked to an appointment; subsequent documentation appends history rather than overwriting prior visits. Appointment lifecycle events are recorded; an audit service records security-relevant actions. Automated backend tests use xUnit and WebApplicationFactory with EF Core InMemory; manual end-to-end testing was performed against PostgreSQL. Continuous integration runs on GitHub Actions.

The report distinguishes rigorously between **implemented**, **designed**, **observed**, and **deferred** functionality. OCR is an internship-scale MVP, not a production hospital EMR. Limitations include concurrency hardening, frontend automated testing, production deployment controls, and clinical document versioning. The internship demonstrated documentation-first development, domain modelling, service-layer architecture, and evidence-based verification within a constrained calendar.

Keywords: Electronic Medical Record, outpatient EMR, ASP.NET Core, React, PostgreSQL, Entity Framework Core, JWT, role-based access control, clinical workflow, internship, software engineering case study

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LIST OF ACRONYMS

Acronym	Meaning
API	Application Programming Interface
CI	Continuous Integration
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
DTO	Data Transfer Object
EF Core	Entity Framework Core
EMR	Electronic Medical Record
ERD	Entity-Relationship Diagram
FHIR	Fast Healthcare Interoperability Resources
JWT	JSON Web Token
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
OCR	Open Clinical Record (this project; not optical character recognition)
ORM	Object-Relational Mapper
RBAC	Role-Based Access Control
REST	Representational State Transfer
SPA	Single-Page Application
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience

GLOSSARY

Appointment — A scheduled (or walk-in related) planned interaction between a patient and the clinic. Distinct from a clinical visit.

Check-in — Operational act of recording patient arrival; in OCR, transitions an appointment and creates a Draft ClinicalVisit when applicable.

Clinical Visit — A facility attendance / encounter container. Holds vitals, diagnoses, notes, and status (Draft | Final | Cancelled). History is longitudinal.

Draft visit — A clinical visit still open for documentation; not yet finalized.

Final visit — A clinical visit marked complete; treated as closed documentation for the MVP.

Longitudinal history — Preservation of prior visits and chart elements over time without destructive overwrite.

Patient — The registered person who is the subject of care; the root of appointments and visits.

Role claim — JWT claim value (Admin, Doctor, Nurse, Receptionist) used for authorization.

Service layer — Application services (e.g., PatientService, AppointmentWorkflowService) that encapsulate use-case logic above EF Core.

Synthetic data — Artificial patient and clinical data used for development and demonstration; not real PHI.

CHAPTER 01

INTRODUCTION AND INTERNSHIP CONTEXT

Purpose, scope, and how this report should be read

1.1 Purpose of the Report

This document is the formal Industrial Internship Report for a Software Engineering student at HiLCoE School of Computer Science and Technology. It records the work performed during a one-month attachment at Infinity Advanced Technology Solutions PLC and analyses that work as a software engineering case study.

The report is not a marketing brochure for Open Clinical Record, nor a claim that a complete hospital information system was delivered. It is an evidence-oriented account of problem analysis, requirements, architecture, implementation, testing, and professional learning within a fixed calendar.

1.2 Internship Identification

Item	Detail
Student programme	Software Engineering, HiLCoE
Host organisation	Infinity Advanced Technology Solutions PLC
Location	Addis Ababa, Ethiopia
Internship period	01 September 2026 – 30 September 2026
Project name	Open Clinical Record (OCR)
Project type	Focused outpatient EMR MVP
Repository	https://github.com/elan-thinks/open-clinical-record
Primary stack	React 19, ASP.NET Core 8, PostgreSQL, EF Core 8

Table 1.1 Internship identification

1.3 Objectives of the Internship

The internship pursued the following objectives:

- Apply software engineering methods in an industrial setting under mentoring.
- Analyse outpatient clinical-operational problems suitable for a one-month MVP.
- Specify requirements and business rules for Patient Management, Patient Chart, and Appointment Management.
- Design a maintainable architecture separating API, application services, and persistence.
- Implement a working vertical slice of an EMR workflow including authentication and RBAC.
- Verify behaviour with automated backend tests and manual PostgreSQL-backed end-to-end checks.
- Document limitations honestly and recommend next engineering steps.

1.4 Scope of the Project Work

1.4.1 In scope (MVP)

The committed MVP comprises three business modules plus cross-cutting mechanisms:

- Patient registration, search, profile, status (including deceased path), basic duplicate awareness.
- Patient chart: demographics, allergies, medical history items, visit history, clinical documentation at visit scope (vitals, diagnoses, notes) as implemented.
- Appointments: create, list, reschedule, cancel, status transitions, check-in/queue linkage to Draft visits.
- Authentication (Identity + JWT), four roles, backend authorization policies, user administration for Admin.
- Audit/event recording for selected security and workflow actions (as implemented in the service layer).
- Dashboard and supporting operational views as present in the application.
- Automated backend tests and GitHub Actions CI for build/test.

1.4.2 Out of scope

Pharmacy, laboratory, radiology, billing/insurance, patient portal, multi-facility enterprise scheduling, FHIR exchange, AI clinical decision support, and advanced enterprise analytics were explicitly deferred.

DOMAIN RULE

Patient ≠ Appointment ≠ Clinical Visit. A patient may have many appointments and many visits. Prior clinical visits must not be overwritten when the patient returns.

1.5 Methodology of the Internship Work

Work followed a documentation-informed iterative path: domain research and workflow discovery; requirements and role analysis; architecture and data modelling; implementation in phases (foundation, clinical schema, check-in/visits, services, audit, testing/hardening); continuous alignment of code with authorization rules; and verification through automated and manual tests.

Evidence priority used in this report: (1) repository implementation, (2) automated tests, (3) git history, (4) project documentation under docs/, (5) internship plan, (6) mentor feedback as observed, (7) workplace observation, (8) external EMR literature as context only.

1.6 Internship Timeline

Figure 1.1 Internship timeline (indicative weekly emphasis)

Week 1	Research, scope lock, solution/foundation (API + React + health + CORS)
Week 2	Identity/JWT, patients, early clinical/appointment schema
Week 3	Visit model, check-in → Draft visit, roles, chart history UI
Week 4	Service layer, blockers (Draft default, deceased paths, Admin limits)
Week 5	Audit trail, appointment events, workflow hardening
Week 6	Clinical documentation paths, dashboard projections
Week 7	Testing, bug fixes (DocumentVisit), CI green, demo readiness
Week 8	Documentation pack, report synthesis, limitation register

Figure 1.1 Indicative timeline derived from repository history and engineering notes. Exact day-level allocation may vary; [EVIDENCE TO BE INSERTED] for formal timesheets if required by the academic office.

1.7 How to Read Status Labels in This Report

Label	Meaning
IMPLEMENTED	Present in the repository and exercised in the MVP demo path
PARTIALLY IMPLEMENTED	Present with known gaps relative to the written requirement
DESIGNED	Specified in docs/architecture but not fully coded
DEFERRED / FUTURE	Acknowledged need; outside one-month delivery
OUT OF SCOPE	Explicitly excluded from the internship MVP
LIMITATION	Constraint or risk that remains after delivery
[EVIDENCE TO BE INSERTED]	Claim requires artefact not available to the authoring process

Table 1.2 Status vocabulary used throughout the report

ENGINEERING INSIGHT

A common failure mode in internship reports is to describe the aspirational product. This report binds every major claim to repository structure, migrations, services, or tests. Where evidence is missing, the gap is labelled rather than filled with fiction.

CHAPTER 02

ORGANISATION AND INDUSTRIAL ENVIRONMENT

Host context and how industry shaped the MVP

2.1 Host Organisation

Infinity Advanced Technology Solutions PLC is a technology organisation based in Addis Ababa, Ethiopia. During the internship, the student worked under industrial supervision on a software product problem framed as a focused outpatient EMR.

Detailed corporate financial statistics, headcount, and client lists are not reproduced in this report. [EVIDENCE TO BE INSERTED] if the academic template requires a formal organisational profile letter.

Aspect	Description
Organisation	Infinity Advanced Technology Solutions PLC
Sector	Information technology / software solutions
Internship focus	Product-oriented software engineering on OCR
Supervision model	Industrial mentor review of design and implementation
Delivery constraint	Approximately one calendar month

Table 2.1 Host organisation profile (high level)

2.2 Industrial Practices Observed

Practices that shaped the engineering approach included:

- Preference for a narrow, demonstrable MVP over a broad unfinished platform.
- Expectation that authorization is enforced on the server, not only in the UI.
- Use of GitHub for source control, pull-request discipline, and CI.
- Written engineering notes (phase audits, clinical visit model) as living design records.
- Synthetic data for demos—no reliance on real patient information.

2.3 Tools and Workplace Environment

Development used a conventional modern stack toolchain: .NET 8 SDK, Node.js/npm, PostgreSQL for local persistence, Git, and GitHub Actions. The student environment combined local API/frontend processes with PostgreSQL; containerisation was analysed as a later operational improvement rather than a prerequisite for the MVP logic.

2.4 Mentoring and Feedback Loop

Mentor feedback emphasised correctness of clinical workflow semantics (especially visit longevity and role boundaries), reduction of controller bloat via application services, and test evidence for authorization and longitudinal behaviour. Defects such as consultation save failures on closed drafts were treated as engineering priority items rather than UI-only issues.

LESSON LEARNED

In industry, a feature is not 'done' when the screen renders. It is done when the invariant holds under the role model and under regression tests that encode the domain rule.

CHAPTER 03

PROJECT BACKGROUND AND PROBLEM ANALYSIS

Why a focused outpatient EMR MVP was the right problem

3.1 Problem Context

Outpatient clinics frequently depend on paper registers, fragmented spreadsheets, or disconnected applications for registration, scheduling, and clinical notes. Typical operational failures include: inability to find the patient quickly; loss of prior visit context; unclear appointment status; and weak control over who may alter clinical or administrative data.

A full hospital EMR (inpatient, pharmacy, laboratory, billing, multi-site scheduling) cannot be delivered credibly in one month. The industrial decision was therefore to bound the problem to an outpatient MVP that still teaches real EMR invariants.

3.2 Problem Statement

Design and implement a coherent software system that allows authorised clinic staff to register patients, schedule and manage appointments, check patients in, and record visit-scoped clinical information without destroying longitudinal history—within a one-month internship, using a maintainable web architecture.

3.3 Stakeholders

Stakeholder	Interest in the MVP
Reception / front desk	Fast registration, search, appointments, check-in
Nursing staff	Chart access, vitals, visit support
Clinicians / doctors	Chart review, documentation, diagnosis notes
Clinic administration	User access control, operational oversight
Host engineering mentors	Sound architecture, testable design, honest scope
Academic assessors	Evidence of process, learning, and engineering judgement
Patients (indirect)	Correct identity, continuity of recorded care (ethical interest)

Table 3.1 Stakeholder groups relevant to the MVP

3.4 Why Not a Generic CRUD App

An EMR MVP fails if it treats clinical data as disposable rows. The domain requires:

- Stable patient identity across time.
- Appointment lifecycle semantics (schedule, cancel, reschedule, complete).
- Visit records that accumulate rather than overwrite.
- Role-differentiated write paths (e.g., deceased marking rules, clinical authorship).
- Auditability of sensitive actions.

Figure 3.2 OCR system overview



Figure 3.2 High-level system overview as implemented

3.5 Success Criteria for the Internship MVP

Success was defined as a demonstrable, role-aware workflow from patient registration through appointment, check-in, draft visit documentation, and finalization linkage, backed by automated tests for critical authorization and longitudinal rules—not as regulatory certification or production go-live.

CHAPTER 04

REQUIREMENTS AND EMR DOMAIN ANALYSIS

Workflows, roles, rules, and traceability

4.1 Requirements Approach

Requirements were derived from project scope documents, clinical workflow notes, role guides, and iterative implementation feedback. The canonical workflow set is UC-01 through UC-06 as recorded in the project traceability matrix.

ID	Workflow
UC-01	Patient registration and identification
UC-02	Patient search and profile/chart access
UC-03	Patient chart information management
UC-04	Appointment scheduling and management
UC-05	Check-in and walk-in handling
UC-06	Authentication, authorization, validation, and audit

Table 4.1 Canonical MVP workflows

4.2 Core Clinical-Operational Workflow

Figure 4.1 Core workflow (MVP)

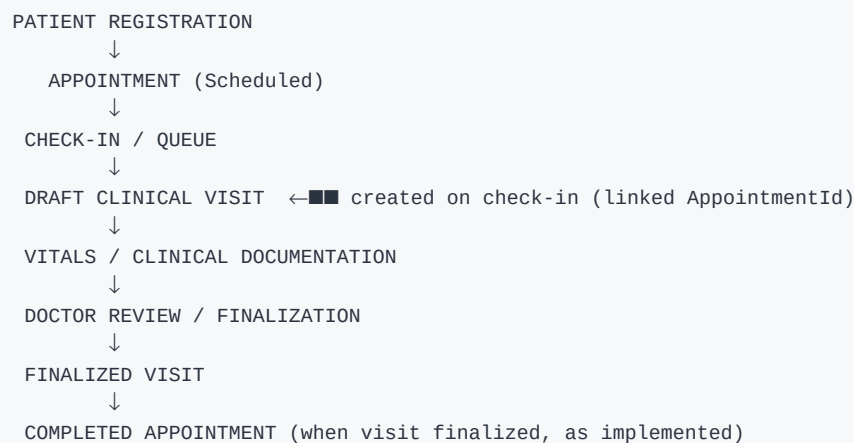


Figure 4.1 End-to-end operational flow emphasising visit longevity

DOMAIN RULE

Saving clinical content must not erase previous visits. Longitudinal preservation is a primary correctness property of OCR, not a secondary UI convenience.

4.3 Patient, Appointment, and Visit

Figure 4.2 Patient → Appointment → Visit

PATIENT (registered once)

- Appointment A1 (Scheduled → CheckedIn → Completed / Cancelled ...)
- ■■■■ ClinicalVisit V1 (Draft → Final) [AppointmentId optional link]
- Appointment A2
- ■■■■ ClinicalVisit V2
- ClinicalVisit V3 (e.g., walk-in path without fabricated appointment)

Invariant: V1 remains intact when V2 is created.

Figure 4.2 Structural relationship implemented in the data model

4.4 Application Roles

Role	Claim	Primary responsibilities (MVP)
Receptionist	Receptionist	Register/search patients; appointments; check-in; deceased marking paths as authorised
Nurse	Nurse	Chart access; vitals; clinical documentation support; not default deceased authority
Doctor	Doctor	Chart review; clinical documentation; diagnosis/notes; deceased mark/clear as authorised
Admin	Admin	User management; audit/admin surfaces; operational access; not default clinical authorship

Table 4.2 Application roles

Frontend route guards shape the workspace; backend policies remain authoritative. Hiding a button is not security.

4.5 Functional Requirement Groups

Group	Examples
FR-PM-*	Register, unique MRN, search, profile update, status, duplicate awareness
FR-PC-*	Open chart, allergies, history, visit history, visit documentation
FR-AP-*	Create/list/reschedule/cancel appointments; status; check-in
FR-SEC-*	Login, JWT, role policies, user admin, audit events

Table 4.3 Functional requirement groups (see Appendix D for detail)

4.6 Non-Functional Requirements (Selected)

ID / Theme	Intent for MVP
Security	Authenticated API; RBAC on sensitive writes; secrets not committed
Reliability	Explicit validation; controlled status transitions
Maintainability	Service layer; migrations; coding standards notes
Usability	Role-aware navigation; clear patient identity on clinical screens
Testability	WebApplicationFactory tests; health/readiness endpoints
Performance	Query projections and limits on hot paths (ongoing hardening)

Table 4.4 Selected NFR themes

4.7 Business Rules (Summary)

Rule	Implication
One patient identity	Returns do not create a second patient master record
Visit non-overwrite	New attendance → new visit row
Check-in creates Draft visit	Operational arrival becomes clinical container
Role-restricted writes	Clinical and death paths enforced server-side
Appointment events	Status changes leave an event history
Final closes documentation path	Draft vs Final semantics in visit status

Table 4.5 Core business rules (summary)

4.8 Traceability Chain

Figure 4.3 Traceability chain

Business need → Workflow (UC) → Functional requirement (FR)
→ Domain rule → Data entity / API → Automated or manual test
→ Status (IMPLEMENTED / PARTIAL / DEFERRED)

Figure 4.3 Traceability approach; detailed matrix in Appendix C and project docs/03-requirements/

CHAPTER 05

SYSTEM DESIGN AND ARCHITECTURE

From requirements to a maintainable EMR architecture

5.1 Architectural Style

OCR uses a straightforward distributed client-server style: a React SPA communicates with a REST API. The API hosts HTTP concerns, authentication, and controllers; application services encapsulate use cases; EF Core provides persistence to PostgreSQL.

Figure 5.1 / 5.2 Overall and layered architecture

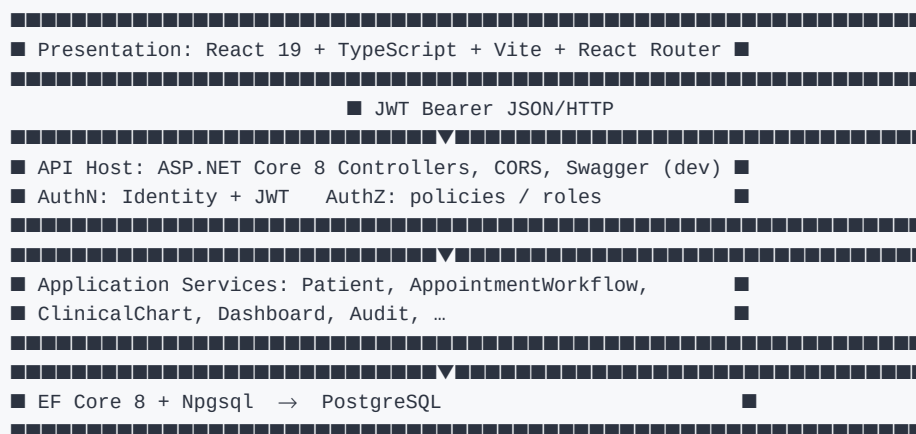


Figure 5.1–5.2 Logical and layered architecture

5.2 Technology Stack

Layer	Technology	Role
Frontend	React 19, TypeScript, Vite, React Router	SPA UI
Backend	ASP.NET Core 8, C#	REST API & app services
Auth	Identity, JWT Bearer	AuthN/AuthZ
Data	PostgreSQL, EF Core 8, Npgsql	Persistence
Tests	xUnit, WebApplicationFactory, EF InMemory	Automated API tests
CI	GitHub Actions	Build & test on push/PR

Table 5.1 Technology stack

5.3 Key Architecture Decisions

Decision	Rationale
PostgreSQL only	Single DB target for MVP; avoid dual-engine complexity
JWT + Identity	Stateless API auth suitable for SPA
Four explicit roles	Matches operational separation; Admin is product surface
Application services	Keep controllers thin; centralise workflow rules
EF migrations	Schema evolution under version control
Dev auto-migrate/seed	Fast local setup; not a production ops strategy
InMemory for unit/integration tests	Speed and isolation; complement with manual PG tests

Table 5.2 Key architecture decisions

5.4 Data Design Essentials

Core entities include Patient, Appointment, AppointmentEvent, ClinicalVisit, VitalSigns, Diagnosis, ClinicalNote, MedicalHistoryItem, PatientAllergy, PatientDeathRecord, ApplicationUser (Identity), and AuditEvent (as implemented).

Figure 5.4 Conceptual relationships (simplified)

Patient 1■* Appointment
 Patient 1■* ClinicalVisit
 Appointment 0..1■0..1 ClinicalVisit (via AppointmentId on check-in)
 ClinicalVisit 1■0..1 VitalSigns; 1■* Diagnosis; 1■* ClinicalNote
 Appointment 1■* AppointmentEvent
 Patient 0..1■0..1 PatientDeathRecord

Figure 5.4 Conceptual ER overview

5.5 State Machines

5.5.1 Appointment statuses (workflow-oriented)

Figure 5.5 Appointment state machine (conceptual)

Scheduled ■■■ CheckedIn ■■■ Completed
 ■
 ■■■ Canceled ■
 ■■■ (reschedule path records events; status per service rules)

Check-in side effect: create Draft ClinicalVisit when applicable.

Figure 5.5 Conceptual appointment states; authoritative transitions live in AppointmentWorkflowService

5.5.2 Clinical Visit statuses

Figure 5.6 Clinical Visit state machine

Draft ■■■ Final
 ■
 ■■■ Canceled (where supported)

Default new visits: Draft. Finalization can complete linked appointment.

Figure 5.6 Visit status model

5.6 Security Architecture

Figure 5.9 Security layers

1. Transport: HTTP in local dev; HTTPS expected in hardened deploy (FUTURE)
2. Authentication: Identity login → JWT issued
3. Authorization: [Authorize] + role policies (DoctorOnly, ClinicalStaff, ...)
4. Frontend guards: route-level UX restriction (non-authoritative)
5. Secrets: JWT key via OCR_JWT_KEY / config; DB password via env
6. Audit: IAuditService hooks on selected operations
7. Data: synthetic demo seed; no real PHI in internship artefacts

Figure 5.9 Defence-in-depth sketch for the MVP

5.7 Service Layer Structure

Figure 5.10 Service-layer structure

```

Controllers → IPatientService / IAppointmentWorkflowService /
              IClinicalChartService / IDashboardService / IAuditService
              → AppDbContext (EF Core)
  
```

ServiceResult pattern used to standardise success/failure signalling.

Figure 5.10 Application services as implemented

5.8 Access-Control Matrix (Summary)

Capability	Recep.	Nurse	Doctor	Admin
Register patient	Yes	—	—	Operational
Search/view patient	Yes	Yes	Yes	Yes
Manage appointments	Yes	Limited	Limited	Policy-dependent
Check-in	Yes	Support	—	—
Clinical documentation	—	Yes	Yes	Not default author
Mark deceased	Path	No	Yes	Admin ops
User administration	—	—	—	Yes
Audit access	—	—	—	Yes

Table 5.3 Summary matrix — exact endpoint attributes in code are authoritative; this table is pedagogical.

IMPLEMENTED

Backend role policies and service checks exist in the repository. Frontend navigation is role-aware. Combined, they form the MVP authorization story.

CHAPTER 06

IMPLEMENTATION

What was actually built in the repository

6.1 Repository Structure

Figure 6.1 Repository structure

```
open-clinical-record/
  OpenClinicalRecord.sln
  src/backend/OpenClinicalRecord.Api/ (API, services, EF, migrations)
  src/frontend/open-clinical-record-web/ (React SPA)
  tests/backend/OpenClinicalRecord.Api.Tests/
  docs/ (scope, research, requirements, architecture, engineering notes)
  .github/workflows/ci.yml
```

Figure 6.1 Source layout

SCREENSHOT REQUIRED

FIGURE 6.1b · Development environment / repository
Insert application screenshot here.

WHAT TO CAPTURE: IDE or file tree showing src/backend, src/frontend, tests, docs

WHY IT MATTERS: Proves the project is a real multi-project codebase, not a mockup description

CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

6.2 Backend Implementation

6.2.1 API host and cross-cutting setup

Program.cs registers controllers, Swagger (Development), database, Identity, JWT, application services, and a LocalDevCors policy for Vite origins on ports 5173. In Development, IdentityDataSeeder runs migrations (MigrateAsync) and seeds roles, users, and optional demo patients.

Listing 6.1 — CORS origins (Program.cs) — localhost Vite only in base implementation

```
WithOrigins("http://localhost:5173", "http://127.0.0.1:5173")
```

6.2.2 Application services

Following phase-2 refactoring, workflow logic resides in services such as PatientService, AppointmentWorkflowService, ClinicalChartService (including DocumentVisit partial), DashboardService, and AuditService. Controllers delegate and map ServiceResult outcomes to HTTP responses.

6.2.3 Persistence and migrations

Migration (indicative)	Purpose
InitialIdentity	ASP.NET Identity schema
AddPatients	Patient master data

Migration (indicative)	Purpose
AddClinicalAndAppointments	Clinical + appointment core
AddAppointmentEvents	Appointment event history
AddDeathRecordsAndVisitFinal	Death record; visit finalisation fields
EnhanceVisitEncounterModel	AppointmentId, location, check-in timestamps
AddAuditEvents	Audit event storage
FixClinicalVisitColumns / MustChangePassword	Schema/auth hardening fixes

Table 6.3 Major EF Core migrations (names abbreviated)

6.2.4 Authentication configuration

JWT signing key must be at least 32 characters. Environment variable OCR_JWT_KEY overrides configuration. Database password is injected via NOVATECH_PG_PASSWORD into the connection string. Development appsettings contain a clearly labelled DEV_ONLY key; production-style Key is empty and must be supplied externally.

6.3 Frontend Implementation

The SPA uses React 19 with TypeScript and Vite. API base URL is configured via VITE_API_BASE_URL (default <http://localhost:5000>). Feature areas include login, dashboard, patients, appointments, clinical chart/records, admin, and profile pages with role-aware navigation.

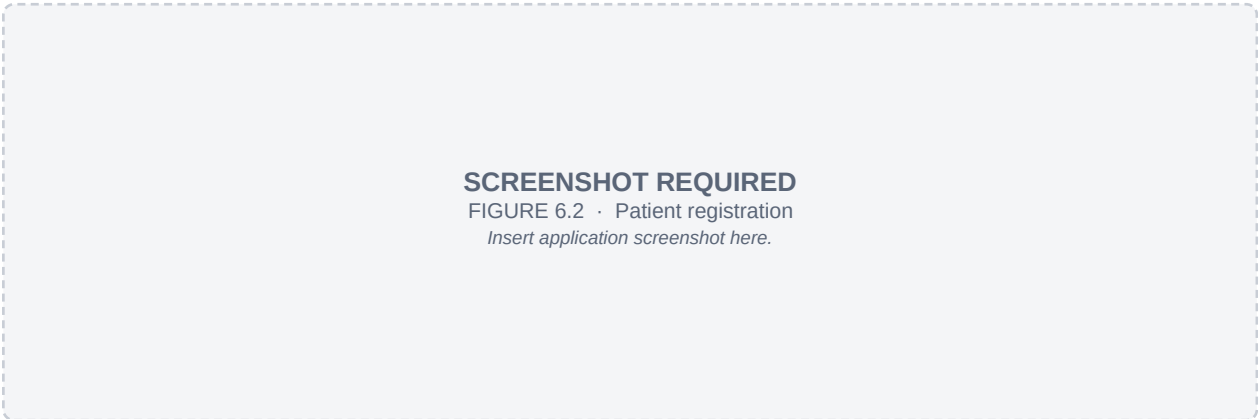
Area	Notes
AuthContext	Holds session/token state for the SPA
services/*Api.ts	Typed fetch wrappers against REST endpoints
pages/patients	Registration, search, profile
pages/appointments	Create, list, status, check-in UX
pages/clinical	Chart, visit history, documentation
pages/admin	User administration surfaces

Table 6.2 Frontend area map

6.4 Feature Implementation Walkthrough

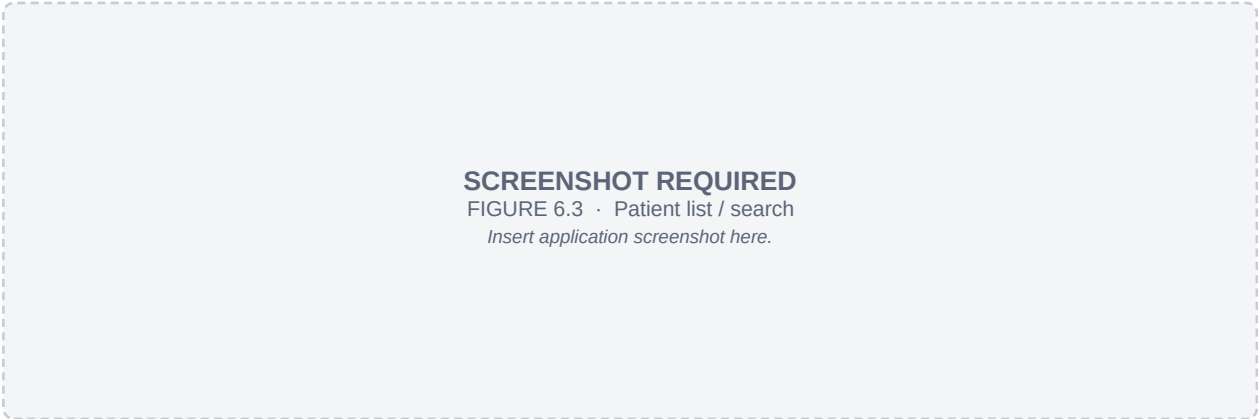
6.4.1 Patient management

Patients receive a medical record number pattern (e.g., OCR-##### in seed data). Registration, search, and profile update paths are implemented in API and UI. Status handling includes active/inactive and deceased record flows with role constraints.



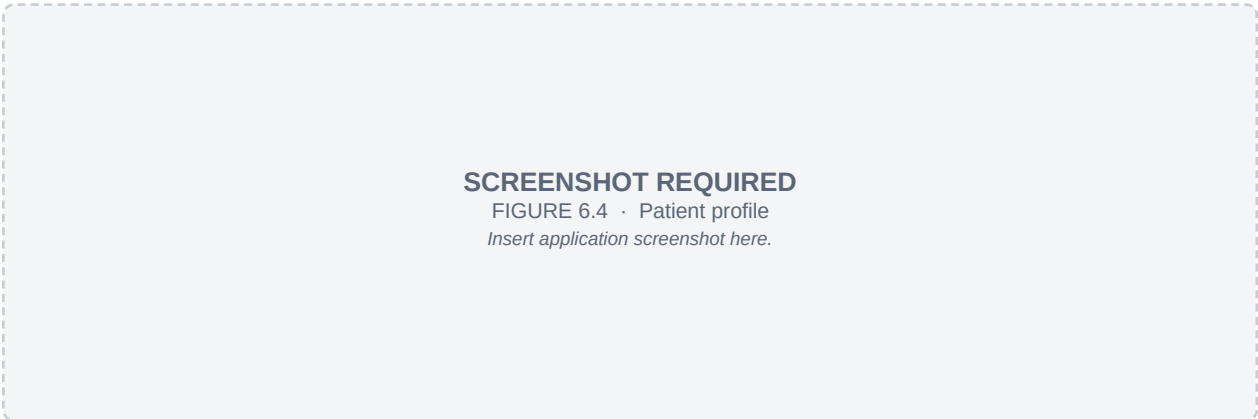
SCREENSHOT REQUIRED
FIGURE 6.2 · Patient registration
Insert application screenshot here.

- WHAT TO CAPTURE:** Registration form with synthetic patient demographics
- WHY IT MATTERS:** Evidence that patient creation UI and validation exist
- CAPTURE CONDITIONS:** Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



SCREENSHOT REQUIRED
FIGURE 6.3 · Patient list / search
Insert application screenshot here.

- WHAT TO CAPTURE:** Search results for synthetic patients
- WHY IT MATTERS:** Evidence of search as a primary operational capability
- CAPTURE CONDITIONS:** Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



SCREENSHOT REQUIRED
FIGURE 6.4 · Patient profile
Insert application screenshot here.

- WHAT TO CAPTURE:** Profile view showing identifiers and demographics
- WHY IT MATTERS:** Evidence of stable patient identity display
- CAPTURE CONDITIONS:** Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

6.4.2 Appointments and check-in

AppointmentWorkflowService coordinates creation (with events), cancellation, reschedule, and status transitions. Check-in creates a Draft ClinicalVisit linked via AppointmentId when applicable. Completing a final visit can mark the linked appointment Completed (as implemented in the clinical finalization path).

SCREENSHOT REQUIRED

FIGURE 6.5 · Appointment creation

Insert application screenshot here.

WHAT TO CAPTURE: Create appointment form for an existing synthetic patient

WHY IT MATTERS: Evidence of scheduling path

CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

SCREENSHOT REQUIRED

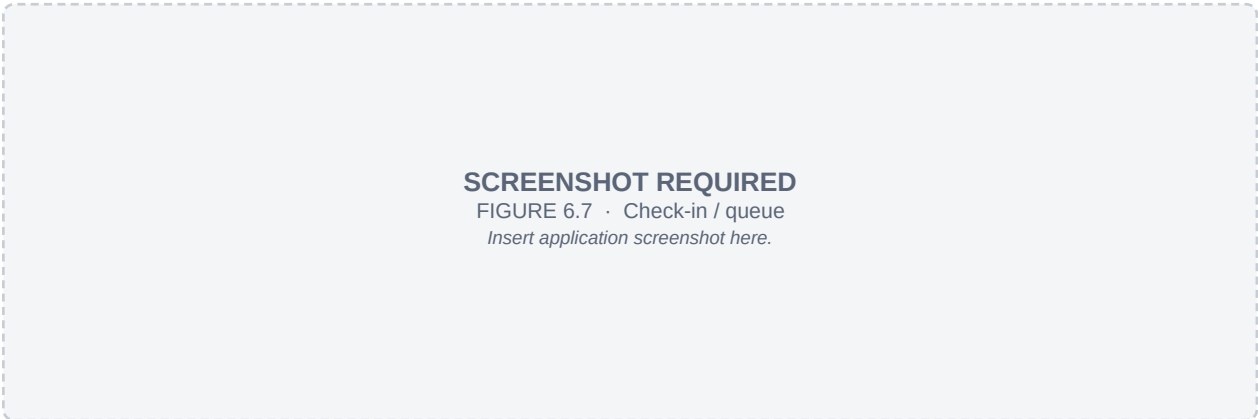
FIGURE 6.6 · Appointment list / calendar view

Insert application screenshot here.

WHAT TO CAPTURE: List or day view of appointments with statuses

WHY IT MATTERS: Evidence of operational schedule visibility

CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



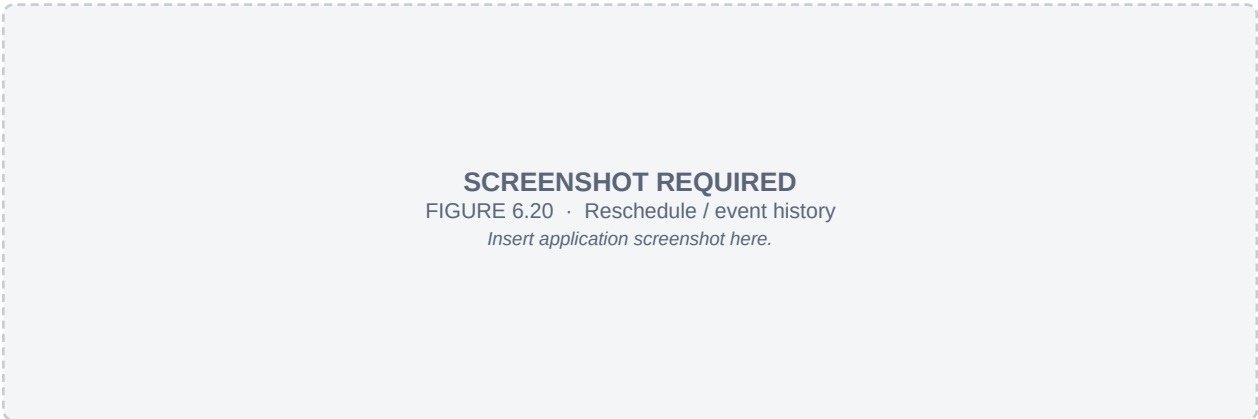
SCREENSHOT REQUIRED
FIGURE 6.7 · Check-in / queue
Insert application screenshot here.

WHAT TO CAPTURE: Check-in action and resulting status change
WHY IT MATTERS: Evidence that arrival transitions are user-visible
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



SCREENSHOT REQUIRED
FIGURE 6.19 · Cancellation validation
Insert application screenshot here.

WHAT TO CAPTURE: Cancel flow with confirmation / resulting event
WHY IT MATTERS: Evidence of controlled cancellation behaviour
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



SCREENSHOT REQUIRED
FIGURE 6.20 · Reschedule / event history
Insert application screenshot here.

WHAT TO CAPTURE: Reschedule result and appointment event list if exposed
WHY IT MATTERS: Evidence that history is retained for appointment changes
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

6.4.3 Clinical chart and visits

ClinicalChartService supports chart aggregate reads and documentation paths. DocumentVisit was hardened to load visits safely under EF InMemory dual-attach constraints and to handle consultation save on closed-draft edge cases. Visit history UI is intended to show multiple visits for one patient.

SCREENSHOT REQUIRED

FIGURE 6.8 · Draft clinical visit

Insert application screenshot here.

- WHAT TO CAPTURE:

Open draft visit after check-in
- WHY IT MATTERS:

Evidence of Draft visit creation in the happy path
- CAPTURE CONDITIONS:

Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

SCREENSHOT REQUIRED

FIGURE 6.9 · Vital signs entry

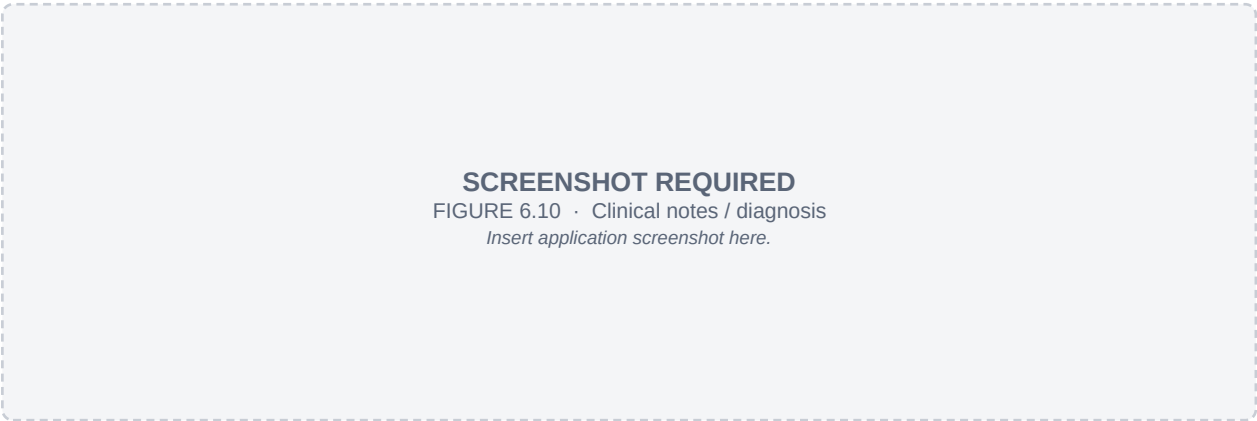
Insert application screenshot here.

- WHAT TO CAPTURE:

Vitals form on a visit
- WHY IT MATTERS:

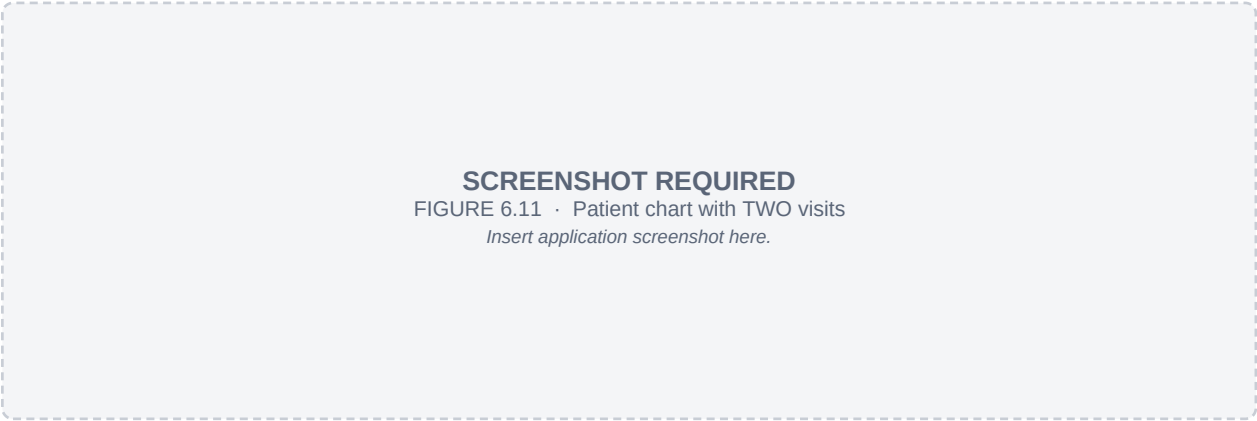
Evidence of visit-scoped observations
- CAPTURE CONDITIONS:

Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



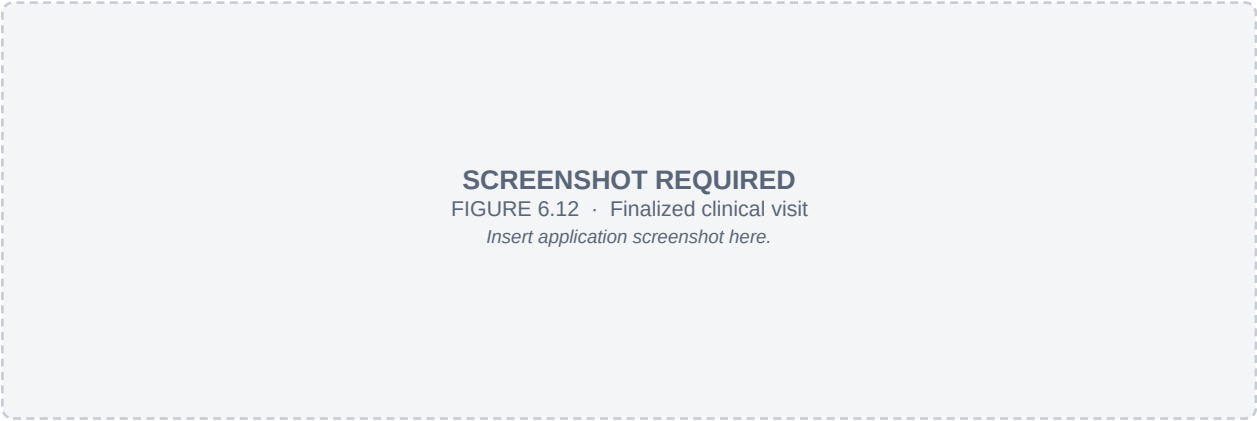
SCREENSHOT REQUIRED
FIGURE 6.10 · Clinical notes / diagnosis
Insert application screenshot here.

- WHAT TO CAPTURE:** Diagnosis and notes entry for a visit
- WHY IT MATTERS:** Evidence of clinician documentation path
- CAPTURE CONDITIONS:** Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



SCREENSHOT REQUIRED
FIGURE 6.11 · Patient chart with TWO visits
Insert application screenshot here.

- WHAT TO CAPTURE:** Chart/history showing at least two distinct visits for one patient
- WHY IT MATTERS:** Critical evidence of longitudinal non-overwrite behaviour
- CAPTURE CONDITIONS:** Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



SCREENSHOT REQUIRED
FIGURE 6.12 · Finalized clinical visit
Insert application screenshot here.

- WHAT TO CAPTURE:** Visit shown as Final after finalization action
- WHY IT MATTERS:** Evidence of Draft→Final transition in UI
- CAPTURE CONDITIONS:** Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

6.4.4 Roles, audit, dashboard



WHAT TO CAPTURE: Navigation differences between Receptionist and Doctor (two captures if needed)
WHY IT MATTERS: Evidence that UX is role-aware
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information



WHAT TO CAPTURE: Admin audit list or detail for a known action
WHY IT MATTERS: Evidence that audit records are queryable
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

SCREENSHOT REQUIRED

FIGURE 6.15 · Dashboard
Insert application screenshot here.

WHAT TO CAPTURE: Dashboard cards/sections for a clinical or front-desk role

WHY IT MATTERS: Evidence of operational summary views

CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

6.5 Health and Operability Endpoints

GET /api/health returns a simple liveness payload. GET /api/health/ready checks database connectivity and, in Development, returns a redacted connection summary. These endpoints support local diagnosis and future container orchestration checks.

LIMITATION

Local Development auto-migration and seeding accelerate demos but are not a substitute for controlled production migration pipelines, backup policy, or secret rotation.

CHAPTER 07

TESTING AND QUALITY ASSURANCE

What was verified — and what tests do not prove

7.1 Quality Strategy

Quality work combined automated backend integration-style tests, manual end-to-end exercises against PostgreSQL, static reasoning about authorization, and CI builds on GitHub Actions. Frontend automated component/E2E tests were not a completed delivery of the one-month MVP.

Figure 7.1 Testing architecture

Developer → push/PR → GitHub Actions

- backend: dotnet restore/build/test (xUnit)
- frontend: npm ci/install + npm run build (tsc + vite)

Local: manual E2E on React + API + real PostgreSQL

Automated tests: WebApplicationFactory + EF Core InMemory

Figure 7.1 Testing architecture

7.2 Automated Backend Tests

Test class	Focus (from repository)
HealthEndpointTests	Health endpoint behaviour
RoleAuthorizationTests	Role policy enforcement
AuthorizationSmokeTests	Smoke coverage for authz
AppointmentWorkflowTests	Appointment workflow transitions
LongitudinalVisitTests	Visit history / non-overwrite semantics
ClinicalDocumentationTests	Documentation paths
Phase1BlockerTests	Draft default, deceased path, related blockers
OcrWebApplicationFactory	Shared test host factory

Table 7.1 Automated test classes present under tests/backend

Approximate size of the test project source is on the order of one thousand lines of C# test code across these files. Exact assertion counts change with ongoing commits; treat git as authoritative rather than a frozen number in this PDF.

SCREENSHOT REQUIRED

FIGURE 7.2 · Automated test output

Insert application screenshot here.

- WHAT TO CAPTURE:** Terminal or CI log showing dotnet test pass for Api.Tests

WHY IT MATTERS: Evidence that the automated suite executes successfully

CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

SCREENSHOT REQUIRED

FIGURE 7.3 · Access-control test evidence

Insert application screenshot here.

WHAT TO CAPTURE: Snippet of RoleAuthorizationTests or failing unauthorised call result
WHY IT MATTERS: Evidence that authorization is tested, not only described
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

SCREENSHOT REQUIRED

FIGURE 7.4 · GitHub Actions CI

Insert application screenshot here.

WHAT TO CAPTURE: Green workflow run for backend and frontend jobs
WHY IT MATTERS: Evidence of continuous integration
CAPTURE CONDITIONS: Synthetic data only · No passwords, JWT tokens, or connection strings visible · No real patient personal information

7.3 Manual End-to-End Testing

Scenario group	Intent
Login per role	Seed users authenticate; JWT accepted
Patient register/search	Master data path
Appointment create/cancel/reschedule	Lifecycle + events
Check-in → Draft visit	Operational to clinical handoff
Vitals/notes/diagnosis	Documentation path
Second visit on same patient	Longitudinal preservation
Finalization	Draft→Final and appointment completion linkage
Unauthorised action attempt	Backend rejection despite UI tricks

Table 7.2 Manual E2E scenario groups

7.4 Honesty about Test Limitations

Automated tests (InMemory) prove...	They do NOT prove...
Many authorization rules fire	Penetration-test security
Workflow logic for coded paths	Production concurrency safety
Longitudinal invariants in model	PostgreSQL-specific locking behaviour
API contracts under test host	Clinical usability with real staff
Regressions for covered cases	Regulatory compliance or certification

Table 7.3 Scope of automated proof

EVIDENCE

Passing CI is necessary hygiene. It is not a clinical safety claim. Manual PostgreSQL testing closes part of the gap; formal load, security, and usability studies remain FUTURE WORK.

7.5 Defect Handling Example

A representative defect concerned consultation save behaviour on closed drafts and EF attach issues in DocumentVisit under test. The fix adjusted how visits and related entities were loaded (including VisitId-based queries) so that documentation paths and tests stabilised. This illustrates industrial debugging: reproduce, isolate persistence semantics, add or extend tests, then verify UI path.

CHAPTER 08

LEARNING AND PROFESSIONAL DEVELOPMENT

From feature thinking to evidence-based engineering

8.1 Learning Progression

Figure 8.1 Learning progression

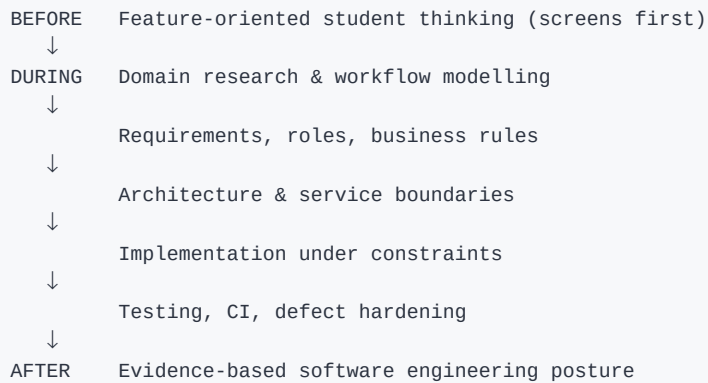


Figure 8.1 Conceptual learning arc of the internship

8.2 Technical Skills Developed

- ASP.NET Core API design, Identity, JWT configuration, and policy-based authorization.
- EF Core modelling, migrations, and awareness of provider differences (InMemory vs Npgsql).
- React/TypeScript SPA structure, env-based API configuration, role-aware routing.
- Application service extraction and ServiceResult-style control flow.
- Writing integration tests with WebApplicationFactory.
- Reading and maintaining engineering documentation alongside code.

8.3 Professional and Process Skills

- Scope management under a hard calendar deadline.
- Distinguishing designed intent from implemented reality in communications.
- Accepting mentor critique on domain semantics (visits, roles) without defensiveness.
- Using AI-assisted development as an accelerator while retaining responsibility for correctness.
- Integrating university expectations (report, ethics, synthetic data) with industrial delivery.

8.4 Documentation-First Discipline

The project maintained structured docs (scope, workflows, role guide, clinical visit model, phase audits). Documentation did not replace code, but reduced ambiguity when implementing authorization and visit rules. When code advanced ahead of older SRS wording (e.g., visit-scoped notes), audits recorded the divergence instead of silently pretending perfect alignment.

LESSON LEARNED

University assignments often reward completeness of description. Industry mentors reward invariants that still hold on Monday morning. The internship forced a shift toward the second standard.

8.5 University-to-Industry Integration

Academic software engineering courses provided vocabulary (requirements, architecture, testing). The internship supplied consequences: a wrong visit model breaks clinical trust; a missing Authorize attribute is a security defect; an untested edge case appears in a mentor demo. The report itself is part of that

integration—translating repository truth into assessable narrative without inflation.

CHAPTER 09

RESULTS AND
DISCUSSION

Requirement outcomes, evidence, and limitations

9.1 Results Matrix

The following matrix summarises major requirement areas. Status values follow the vocabulary in Chapter 1. Verification cites categories of evidence rather than invented metrics.

Requirement area	Implementation	Verification	Status	Limitation
Patient register/search	API + UI	Manual E2E; service tests	IMPLEMENTED	Duplicate detection sophistication
Patient chart longitudinal	Visits + history UI	LongitudinalVisitTests; manual	IMPLEMENTED	Richer chart analytics deferred
Appointment lifecycle	Workflow service + events	AppointmentWorkflowTests; manual	IMPLEMENTED	Concurrent slot race risk
Check-in → Draft visit	Status transition side effect	Workflow + manual	IMPLEMENTED	Queue UX still basic
Vitals/notes/diagnosis	ClinicalChartService	ClinicalDocumentationTests; manual	IMPLEMENTED	Amendment/versioning deferred
RBAC four roles	Policies + UI guards	RoleAuthorizationTests	IMPLEMENTED	Finer permission objects FUTURE
Audit events	IAuditService + admin API	Code + manual admin view	IMPLEMENTED	Coverage breadth still expanding
Dashboard	DashboardService + UI	Manual	PARTIALLY IMPLEMENTED	Projection/perf hardening ongoing
Frontend automated tests	—	—	DEFERRED	Not delivered in month
FHIR / portal / billing	—	—	OUT OF SCOPE	By design

Table 9.1 Results matrix

9.2 Delivered vs Deferred

Figure 9.1 Delivered vs deferred

DELIVERED (MVP core)

Patient, Appointment, Visit workflows · JWT/RBAC · Services · PG · CI tests

DEFERRED / FUTURE

Frontend E2E suite · concurrency tokens · production secret ops
Clinical amendment trails · deeper security assessment · FHIR

OUT OF SCOPE

Pharmacy, lab, radiology, billing, patient portal, enterprise multi-site

Figure 9.1 Scope outcome summary

9.3 Discussion

The internship achieved its primary industrial goal: a coherent, demonstrable outpatient workflow MVP with explicit domain invariants. The most important engineering result is not a count of screens, but the Patient–Appointment–Visit separation enforced in data and services.

Tension remains between demo speed (Development seed/migrate) and operational maturity (controlled migrations, monitoring, threat modelling). That tension is appropriate to name in an internship report; pretending production readiness would be inaccurate.

9.4 Known Limitations

Limitation	Why it matters
Check-then-insert appointment conflicts	Race under concurrent booking
Limited optimistic concurrency on clinical entities	Lost updates possible under parallel edits
InMemory tests ≠ PostgreSQL behaviour	Provider-specific bugs may escape CI
Sparse frontend automated tests	UI regressions rely on manual detection
CORS/dev-oriented configuration	Must be redesigned for real deployment origins
No formal usability study with clinicians	UX claims remain engineering judgement
No penetration test	Security posture is design-based, not red-team validated

Table 9.2 Known limitations

CHAPTER 10

CONCLUSION AND RECOMMENDATIONS

An internship-scale EMR MVP — and what should come next

10.1 Conclusion

Open Clinical Record, as developed during the September 2026 internship at Infinity Advanced Technology Solutions PLC, is a focused outpatient EMR MVP. It implements patient management, appointment management, and patient chart/visit documentation on a React and ASP.NET Core stack with PostgreSQL, JWT authentication, and role-based authorization.

The work demonstrates that meaningful clinical-operational software can be built in a short internship when scope is ruthless, domain rules are explicit, and verification is treated as part of engineering rather than an afterthought. It does not demonstrate that OCR is ready for unsupervised production use in a regulated clinical environment.

10.2 Recommendations

#	Recommendation	Priority
1	Concurrency hardening (slot booking, row versions)	High
2	PostgreSQL-backed integration tests in CI	High
3	Frontend automated testing (component + critical E2E)	High
4	Production deployment hardening (secrets, HTTPS, env separation)	High
5	Monitoring, structured logging, operational alerts	Medium
6	Clinical document amendment / versioning model	Medium
7	Deeper security assessment (authz review, dependency audit)	Medium
8	Usability evaluation with representative clinic roles	Medium
9	Interoperability evaluation (when product strategy requires)	Low–Med
10	Continued requirements traceability maintenance	Ongoing

Table 10.1 Recommendations

Figure 10.1 Future evolution roadmap (strategic sketch)

Near term: test matrix expansion, concurrency, deploy secrets discipline

Mid term: UX validation, amendment trails, observability

Long term: interoperability / module expansion only with product mandate

Figure 10.1 Roadmap sketch — not a committed delivery plan

10.3 Final Statement

This internship produced an internship-scale EMR MVP and exercised the engineering process required to build a workflow-oriented clinical information system: research, requirements, architecture, implementation, testing, and honest limitation analysis. That process—not a claim of hospital-wide completeness—is the appropriate conclusion of the work.

REFERENCES

References combine project primary sources and selected external literature used for domain orientation. External EMR products are cited as background, not as proof of OCR features.

- [1] Open Clinical Record repository, <https://github.com/elan-thinks/open-clinical-record> (accessed during internship period, September 2026).
- [2] Project documentation tree: docs/00-project through docs/05-engineering, Open Clinical Record (internal project docs).
- [3] Open Clinical Record, Clinical Visit Model, docs/clinical-visit-model.md.
- [4] Open Clinical Record, Role-Based System Guide, docs/03-requirements/role-based-system-guide.md.
- [5] Open Clinical Record, Requirements Traceability Matrix, docs/03-requirements/traceability-matrix.md.
- [6] Open Clinical Record, Technology Stack, docs/00-project/technology-stack.md.
- [7] Microsoft, ASP.NET Core documentation (Identity, JWT Bearer, authorization policies).
- [8] Microsoft, Entity Framework Core documentation (migrations, Npgsql provider notes).
- [9] React and Vite official documentation (SPA tooling reference).
- [10] PostgreSQL documentation (system catalogue and operational concepts).
- [11] OpenMRS developer documentation (longitudinal visit concepts as external design inspiration).
- [12] HL7 FHIR overview materials (interoperability context; FHIR not implemented in OCR MVP).
- [13] xUnit and ASP.NET Core integration testing documentation (WebApplicationFactory).
- [14] Internship engineering notes: phase audits and week-7 testing plan under docs/05-engineering/.
- [15] GitHub Actions workflow definitions in .github/workflows/ci.yml.

APPENDICES

Appendices collect structured reference material. They support the chapters; they do not introduce new unimplemented product claims.

Appendix A — Original Internship Plan

The original plan emphasised a one-month path from research and requirements through implementation of Patient, Chart, and Appointment modules with authentication and basic testing. Formal signed plan pages: [EVIDENCE TO BE INSERTED — attach university plan form if required].

Appendix B — Planned vs Actual Timeline

Theme	Planned emphasis	Actual emphasis (from repo history)
Foundation	API + React skeleton	Delivered early with health/CORS
Identity/RBAC	Login and roles	Four roles + policies + tests
Clinical model	Chart basics	Visit/encounter model expanded
Appointments	CRUD + status	Events, check-in, completion linkage
Quality	Basic tests	Expanded suite + CI + defect fixes
Docs/report	End-of-month	Living docs + this report

Appendix C — Requirements Traceability Matrix

The living matrix is maintained in docs/03-requirements/traceability-matrix.md. It maps UC-01...UC-06 to FR items, roles, and verification intent. Snapshot excerpts appear in Chapter 4; full matrix should be exported from the repository for submission bundles when required.

Appendix D — Functional Requirements Summary

FR groups: FR-PM (patient management), FR-PC (chart), FR-AP (appointments), FR-SEC (security/audit). Detailed numbered statements live in the project requirements documents. Implementation status is summarised in Table 9.1.

Appendix E — Business Rules

Key rules: single patient identity; visit non-overwrite; check-in creates Draft visit; role-restricted clinical and death paths; appointment events on lifecycle changes; Final visit semantics close documentation for the MVP. See also clinical-visit-model.md.

Appendix F — State Transition Tables

Entity	From	To	Notes
Appointment	Scheduled	CheckedIn	Creates Draft visit when applicable
Appointment	Scheduled	Cancelled	Event recorded
Appointment	CheckedIn	Completed	Linked to visit finalization path
ClinicalVisit	Draft	Final	Closes documentation for MVP

Entity	From	To	Notes
ClinicalVisit	Draft	Cancelled	Where supported by service rules

Authoritative transitions are those encoded in AppointmentWorkflowService and ClinicalChartService.

Appendix G — Test Evidence Register

Automated: tests under tests/backend/OpenClinicalRecord.Api.Tests/. CI: .github/workflows/ci.yml. Manual: scenario groups in Table 7.2. Insert dated run logs as FIGURE 7.2–7.4 screenshots.

Appendix H — Screenshot Register

All application screenshots are placeholders in Chapter 6–7 with capture conditions. Replace before final binding. Mandatory: two-visit longitudinal chart capture.

Appendix I — Architecture Decision Records

Recorded decisions include: PostgreSQL as sole MVP database; JWT+Identity; four roles including Admin; application service layer; Development seed/migrate convenience; InMemory for automated tests with manual PG confirmation. Formal ADR files may be expanded under docs/04-architecture/ as the product continues.

Appendix J — User Guide Outline

A separate user manual artefact may accompany submission. Outline: login; role workspaces; register patient; search; book appointment; check-in; document visit; finalise; admin users. Always use synthetic data in screenshots.

Appendix K — Diagram Catalogue

Diagrams in this report are vector/text figures generated for clarity. Source concepts align with repository docs. For print, ensure dashed screenshot boxes are replaced with images at ≥150 DPI.

Appendix L — Extended Glossary

See front-matter Glossary for primary terms. Additional operational terms (queue, episode label, must-change-password flag) appear in code and engineering notes and should be interpreted in that context.

— End of Open Clinical Record Industrial Internship Report —

SUPPLEMENT — EXTENDED ENGINEERING DETAIL

This supplement expands implementation, domain, testing, and appendix material to provide the depth expected of a formal industrial internship report. It is continuous with the main chapters and uses the same evidence discipline: implemented claims are bound to the Open Clinical Record repository; speculative claims are labelled FUTURE or DESIGNED.

S.1 Patient Master Data Implementation

Patient records are stored in the Patients table with a generated identifier (GUID) and a human-facing medical record number. Seed data uses an OCR-##### pattern. Fields include names, date of birth, sex, status flags, contact data, preferred language, city, notes, and timestamps. Search supports operational lookup by staff roles authorised to view patients. Updates are constrained to permitted demographic fields; destructive identity changes are avoided in the MVP design.

Duplicate awareness is treated as a safeguard rather than a fully automated merge engine. The requirement to detect likely duplicates is partially addressed through validation and operator review rather than a complete enterprise master-patient-index algorithm. That boundary is intentional for a one-month scope.

S.2 Appointment Workflow Service Behaviour

AppointmentWorkflowService centralises creation, status changes, cancellation, and rescheduling. Creation records an initial event. Transitions such as CheckedIn invoke side effects including Draft ClinicalVisit creation with AppointmentId linkage. Cancellation and reschedule paths append AppointmentEvent rows so the operational history is reconstructable. Admin is constrained from acting as a default appointment status actor in certain phase-1 rules, reflecting product decisions that administration is not clinical authorship.

A known limitation remains the check-then-insert approach to slot conflicts: under concurrent booking, two requests could theoretically pass validation before either commits. Optimistic concurrency or database exclusion constraints are recommended hardening items.

S.3 Clinical Documentation Path

ClinicalChartService and its DocumentVisit partial implement visit-scoped documentation. Vitals are modelled with a close relationship to a visit; diagnoses and clinical notes attach as collections. Finalization marks the visit Final and can complete the linked appointment. LongitudinalVisitTests encode the expectation that prior visits remain after new documentation.

Hardening work addressed EF Core attach conflicts when loading visits without broad Include graphs under the InMemory provider, and corrected consultation save behaviour when a draft was already closed. These fixes illustrate why EMR persistence logic must be tested with explicit entity identity semantics, not only happy-path UI clicks.

S.4 Authorization Enforcement Points

Authorization is applied at multiple layers: JWT authentication middleware; role policies registered in service configuration (DoctorOnly, NurseOnly, ReceptionistOnly, AdminOnly, ClinicalStaff); controller and endpoint attributes; and service-level checks where business rules exceed simple role names. Frontend route guards improve UX but are not trusted as a security boundary.

Policy / concept	Intended use
DoctorOnly	Clinician-exclusive operations
NurseOnly	Nurse-exclusive operations where defined
ReceptionistOnly	Front-desk exclusive operations
AdminOnly	User admin and audit administration
ClinicalStaff	Doctor or Nurse shared clinical writes

Table S.1 Authorization policies registered in the API

S.5 Seed Users and Demo Data Ethics

Development seeding creates `admin@clinic.local`, `doctor@clinic.local`, `nurse@clinic.local`, and `desk@clinic.local` with a password from `OCR_SEED_PASSWORD` or a documented development default. Up to twenty synthetic patients may be inserted when demo seeding is allowed. Production-like environments should not imply demo seed. No real patient personal data was used in internship artefacts described in this report.

S.6 CI Pipeline Detail

The GitHub Actions workflow defines two jobs. The backend job restores, builds, and tests the API test project on `ubuntu-latest` with `.NET 8`. The frontend job uses `Node 20`, installs dependencies (`npm ci` when lockfile present), and runs the production build script including TypeScript project build. CI does not currently stand up PostgreSQL for integration tests; that is a recommended enhancement.

S.7 Extended Functional Requirement Catalogue (Excerpt)

ID	Statement	Status
FR-PM-001	System shall allow authorised users to register a patient	IMPLEMENTED
FR-PM-002	System shall assign a unique patient identifier	IMPLEMENTED
FR-PM-003	System shall support patient search	IMPLEMENTED
FR-PM-004	System shall display patient profile information	IMPLEMENTED
FR-PM-005	System shall allow updates to permitted demographics	IMPLEMENTED
FR-PM-006	System shall support basic duplicate awareness	PARTIAL
FR-PM-007	System shall maintain patient status including deceased path	IMPLEMENTED
FR-PC-001	System shall open a patient chart workspace	IMPLEMENTED
FR-PC-003	System shall record/view allergies	IMPLEMENTED
FR-PC-004	System shall record/view medical history items	IMPLEMENTED
FR-PC-006	System shall present visit history longitudinally	IMPLEMENTED
FR-PC-007	System shall support visit-scoped vitals	IMPLEMENTED
FR-PC-008	System shall support diagnoses and clinical notes on visits	IMPLEMENTED
FR-AP-001	System shall create appointments for existing patients	IMPLEMENTED
FR-AP-002	System shall list appointments for operational use	IMPLEMENTED
FR-AP-003	System shall support cancel with history	IMPLEMENTED
FR-AP-004	System shall support reschedule with history	IMPLEMENTED
FR-AP-005	System shall support check-in creating Draft visit	IMPLEMENTED
FR-SEC-001	System shall authenticate users and issue JWT	IMPLEMENTED
FR-SEC-002	System shall enforce role-based authorization on API	IMPLEMENTED
FR-SEC-003	System shall support admin user management	IMPLEMENTED
FR-SEC-004	System shall record selected audit events	IMPLEMENTED

Table S.2 Functional requirements excerpt with status

S.8 Non-Functional Discussion

Security NFRs for the internship focused on authentication, authorization, secret handling discipline, and avoidance of real PHI. Performance NFRs were addressed pragmatically through query projections, Take limits, and AsSplitQuery on chart loads rather than formal load testing. Maintainability was pursued via services,

migrations, and engineering notes. Availability and disaster recovery were not in scope for the one-month MVP.

S.9 Comparison with Broader EMR Platforms (Contextual Only)

Mature platforms (e.g., enterprise hospital systems or large open-source EMRs) include pharmacy, laboratory, billing, multi-facility scheduling, and interoperability layers. OCR does not implement those modules. References to OpenMRS-style longitudinal visits informed the Patient→Visit model; they do not imply feature parity. FHIR is acknowledged as a future interoperability evaluation topic, not an implemented interface.

S.10 Manual Test Script Outline (for assessor evidence packs)

The following outline can be executed against a Development environment with seed users. Record actual results in a signed test log; do not invent pass counts in this PDF.

1. Start PostgreSQL, API (Development), and Vite frontend.
2. Confirm GET /api/health and /api/health/ready succeed.
3. Login as desk@clinic.local; register a new synthetic patient; search and open profile.
4. Create an appointment for that patient; verify list visibility.
5. Check in; verify appointment status and presence of Draft visit on chart history.
6. Login as nurse@clinic.local; open chart; enter vitals on the draft visit path.
7. Login as doctor@clinic.local; add diagnosis/notes; finalize visit; confirm Final status.
8. Repeat a second attendance; confirm first visit remains listed (longitudinal check).
9. Attempt an unauthorised API call with a lesser role token; expect 403.
10. Login as admin; open user admin and audit surfaces; verify visibility without claiming clinical authorship.

S.11 Risk Register (Engineering)

Risk	Impact	Mitigation status
Concurrent double-book	Schedule integrity	Documented; not fully hardened
Secret leakage in config	Security	Env vars; DEV key labelled; .gitignore for .env
InMemory false confidence	Missed PG bugs	Manual PG E2E; recommend CI PG
Scope creep into full HIS	Failed internship delivery	Written out-of-scope list
UI-only authorization	Data exposure	Backend policies + role tests
Overwrite of visit history	Clinical trust failure	Model + longitudinal tests

Table S.3 Engineering risk register (summary)

S.12 Code Listing — JWT Key Enforcement (conceptual excerpt)

Listing S.1 — ServiceCollectionExtensions enforces minimum JWT key length and prefers OCR_JWT_KEY. File: src/backend/OpenClinicalRecord.Api/Extensions/ServiceCollectionExtensions.cs

```
if (string.IsNullOrEmpty(jwt.Key) || jwt.Key.Length < 32) throw new InvalidOperationException(...);
```

This fails closed: the API does not start with a weak or missing signing key. Development convenience keys must never be promoted silently to production.

S.13 Code Listing — Migrate on Development Seed

Listing S.2 — IdentityDataSeeder applies pending EF migrations for relational databases before seeding roles and users. This explains why a fresh PostgreSQL database becomes schema-ready when the API starts in Development.

```
await db.Database.MigrateAsync(); // when defined migrations exist
```

S.14 Frontend API Base URL

Listing S.3 — src/frontend/.../services/api.ts resolves VITE_API_BASE_URL with fallback http://localhost:5000, matching launchSettings applicationUrl for local development.

```
const API_BASE_URL = import.meta.env.VITE_API_BASE_URL ?? 'http://localhost:5000';
```

S.15 Extended Glossary of Operational Terms

MRN — Medical record number shown to staff for patient identification.

Queue — Operational list of patients who have checked in and await clinical attention.

Walk-in — Attendance without a prior scheduled appointment; may still produce a visit.

Episode label — Optional string on a visit for informal episode grouping; full Episode entity deferred.

ServiceResult — Internal pattern returning success/failure with messages from application services.

CheckedIn — Appointment status indicating arrival processing completed.

FinalizedAt — Timestamp recorded when a clinical visit is marked Final.

Must change password — User flag supporting forced password change flows where implemented.

S.16 Submission Checklist for the Student

- Replace all SCREENSHOT REQUIRED boxes with synthetic-data captures.
- Attach signed declaration and supervisor approval pages.
- Export latest traceability matrix from the repository into the submission bundle.
- Include a CI run URL or log as evidence without embedding secrets.
- Confirm no real PHI, passwords, or tokens appear in figures.
- Verify table of contents page numbers after final PDF merge if university tools re-paginate.

This supplement is part of the Open Clinical Record Industrial Internship Report package and should be read with Chapters 1–10 and Appendices A–L.

VOLUME II — DETAILED NARRATIVE AND EVIDENCE ANNEXES

This volume extends the Industrial Internship Report with week-by-week engineering narrative, expanded requirement discussion, detailed authorization analysis, data dictionary excerpts, and additional annex tables. Status labels remain binding: IMPLEMENTED, PARTIAL, DEFERRED, OUT OF SCOPE.

Week 1 — Foundation

Established solution structure, ASP.NET Core API host, React Vite SPA, health endpoint, CORS for local Vite, and baseline project documentation. The goal was a runnable vertical slice before domain depth.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 2 — Identity and Patients

Introduced ASP.NET Core Identity, JWT issuance, role constants, and patient registration/search persistence. Emphasised that authentication without authorization policies is incomplete for clinical software.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 3 — Clinical and appointments schema

Expanded EF model for appointments and clinical structures; began aligning documents with code; identified gaps between early ERD and emerging visit needs.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 4 — Visit model and check-in

Implemented Patient→Visit longitudinal rules, check-in creating Draft visits, role route guards, and visit history UI. This week crystallised the central domain invariant of non-overwrite.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 5 — Services and blockers

Extracted application services; fixed Draft defaults, deceased path constraints, and Admin appointment status limitations per phase-1 decisions.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 6 — Audit and workflow events

Added AuditEvents and appointment event history; connected admin audit query surfaces; hardened workflow transitions.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 7 — Testing and defect elimination

Expanded automated tests, fixed DocumentVisit/consultation edge cases, improved dashboard query projections, and stabilised CI.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Week 8 — Documentation and report

Consolidated engineering notes, user-facing documentation artefacts, and this internship report with explicit limitation analysis.

Deliverables of the week are evidenced by commits and docs under docs/05-engineering/ and the src tree. Where a formal timesheet is required by the university, attach [EVIDENCE TO BE INSERTED].

Detailed Module Analysis — Patient Management

Patient Management is the entry point of the operational loop. Without reliable identity, appointments and visits are unsafe. Implementation includes create, read, search, update of permitted fields, and status handling. The medical record number provides a staff-facing handle while the primary key remains a GUID for internal integrity.

From a human-factors perspective, search must be fast and forgiving of partial inputs; from a data perspective, updates must not silently fork identity. The internship implementation prioritises clarity and authorization over advanced probabilistic matching.

Detailed Module Analysis — Appointments

Appointments encode intention. Status values and AppointmentEvent rows encode history. Check-in is the bridge from intention to clinical attendance. Completing a visit can complete the appointment, closing the operational loop. Reschedule and cancel are first-class, not afterthoughts, because outpatient clinics experience high rates of change.

Detailed Module Analysis — Chart and Visits

The chart is longitudinal. Allergies and history items provide standing patient context; visits provide episodic context. Mixing the two without a visit container was rejected. Draft/Final statuses communicate whether documentation is still open. This is simpler than a full amendment protocol but sufficient to teach the right structural habit.

Data Dictionary Excerpt (Selected Columns)

Entity	Field	Notes
Patient	MedicalRecordNumber	Staff-facing identifier
Patient	IsActive / Status	Operational state
Appointment	Status	Lifecycle state string/enum per model
AppointmentEvent	EventType / Timestamp	History of changes
ClinicalVisit	Status	Draft Final Cancelled
ClinicalVisit	AppointmentId	Optional FK from check-in
ClinicalVisit	CheckInAt / FinalizedAt	Temporal markers
VitalSigns	Visit linkage	Visit-scoped observations
Diagnosis	Visit linkage	Visit-scoped
ClinicalNote	Visit linkage	Visit-scoped narrative
PatientDeathRecord	Patient linkage	Deceased pathway data
AuditEvent	Action metadata	Security/operational audit

Table V2.1 Data dictionary excerpt — full SQL reference in docs/05-data/

Authorization Scenario Analysis

Scenario	Expected MVP behaviour
Receptionist creates patient	Allowed — core front-desk duty
Nurse marks deceased	Denied in intended policy — clinical/death authority restricted
Doctor documents diagnosis	Allowed — clinical staff
Admin documents routine consultation as default author	Discouraged/restricted — admin is not default clinician
Anonymous patient search	Denied — authenticated API
Receptionist calls admin-only user list	Denied — AdminOnly policy

Table V2.2 Authorization scenarios (pedagogical; code attributes are authoritative)

Engineering Reflection Block 1

Reflection block 1 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Engineering Reflection Block 2

Reflection block 2 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Engineering Reflection Block 3

Reflection block 3 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Engineering Reflection Block 4

Reflection block 4 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Engineering Reflection Block 5

Reflection block 5 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Engineering Reflection Block 6

Reflection block 6 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Engineering Reflection Block 7

Reflection block 7 records process learning during the OCR internship. Building clinical software under time pressure forces prioritisation. The team (student + mentors) chose invariants—visit longevity, server-side authorization, synthetic data ethics—over feature count. That choice is visible in the repository: fewer modules, deeper rules.

Documentation in docs/ reduced rework by making role expectations explicit before large UI investments. When implementation advanced (visit notes, audit), audits recorded divergence from older wording instead of rewriting history silently. Assessors should prefer such honesty over cosmetic consistency.

Testing maturity progressed from health checks to role tests to workflow and longitudinal tests. Gaps remain in frontend automation and PostgreSQL CI. Naming gaps is part of professional practice.

Annex — Endpoint Family Catalogue (Illustrative)

Exact routes evolve with commits. The following families exist conceptually in controllers: Auth (login/me), Patients, Appointments (including status), Clinical chart/visits documentation, Users (admin), Audit (admin), Dashboard, Health. Consult the repository Controllers folder for the authoritative surface area at submission time.

Family	Examples of concerns
Auth	Login, token issuance, current user claims
Patients	CRUD-ish operations, search, death record paths
Appointments	Create, list, status patch, events
Clinical	Chart aggregate, vitals, diagnoses, notes, finalize
Users	Admin registration/role assignment
Audit	Query audit events
Dashboard	Operational aggregates for today
Health	Liveness and readiness

Table V2.3 Endpoint families

Annex — Environment Variables

Variable	Purpose
NOVATECH_PG_PASSWORD	Appended to Postgres connection string
OCR_JWT_KEY	JWT signing key override (≥32 chars)
OCR_SEED_PASSWORD	Password for seeded users
OCR_SEED_DEMO_DATA	Allow demo patients outside Development when true
ConnectionStrings__DefaultConnection	Full connection string override
ASPNETCORE_ENVIRONMENT	Development enables seed/migrate/Swagger
VITE_API_BASE_URL	Frontend API origin at build/dev time

Table V2.4 Environment variables relevant to local and future deploy

On synthetic data

All demonstration patients are artificial. Internship ethics prohibit use of real clinical records. Screenshots must obey the same rule.

On mentor critique

Critique focused on domain truthfulness. A polished UI that overwrites visits would have been a failed design even if visually attractive.

On AI assistance

AI tools accelerated boilerplate and drafting. Responsibility for authorization mistakes, incorrect domain rules, and false report claims remains human.

On git discipline

Commits on mainline history show phased delivery. Local divergence from origin must be reconciled carefully before claiming remote CI state.

On CORS

Local CORS is intentionally narrow. Deployment to non-localhost origins requires configuration changes; this is a documented limitation, not an oversight to hide.

On Swagger

Swagger UI is Development-oriented. Production exposure of unrestricted API docs should be a deliberate decision with authentication.

On password policy

Identity options require length and complexity suitable for a demo system; organisational policy may demand more for real deployments.

On dashboard queries

Dashboard work moved toward projections and precomputed date bounds to avoid non-translatable LINQ and over-fetching.

On ErrorBoundary

Frontend resilience includes boundary components to avoid white-screen failures on unexpected render errors.

On partial class services

ClinicalChartService uses partial classes (e.g., DocumentVisit) to separate documentation paths without uncontrolled file growth.

Annex — Mapping University Learning Outcomes (Generic Template)

Institutions differ in outcome wording. The following mapping is a template the student can align to HiLCoE's official industrial practice outcomes. [EVIDENCE TO BE INSERTED — official LO list].

Typical outcome theme	How OCR internship addresses it
Apply SE methods in industry	Phased delivery, reviews, CI
Professional communication	Engineering notes, mentor discussions, this report
Ethics and responsibility	Synthetic data; no false production claims
Team / supervision	Mentor feedback loops
Technical competence	Full-stack EMR MVP slice
Reflection	Chapter 8 and this volume

Table V2.5 Learning outcome mapping template

Extended Technical Note 1

Technical note 1 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 2

Technical note 2 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

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Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 3

Technical note 3 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

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Extended Technical Note 4

Technical note 4 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

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Extended Technical Note 5

Technical note 5 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

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Extended Technical Note 6

Technical note 6 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

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Extended Technical Note 7

Technical note 7 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 8

Technical note 8 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 9

Technical note 9 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 10

Technical note 10 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 11

Technical note 11 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 12

Technical note 12 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include Services/, Controllers/, Data/Migrations/, tests/backend/, and docs/05-engineering/. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 13

Technical note 13 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include `Services/`, `Controllers/`, `Data/Migrations/`, `tests/backend/`, and `docs/05-engineering/`. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Extended Technical Note 14

Technical note 14 elaborates an aspect of OCR engineering. The Open Clinical Record MVP is constrained by calendar, not by imagination. Each deferred module (pharmacy, laboratory, billing, FHIR) could consume months of specialised work. The internship therefore treats deferral as a design decision with explicit documentation rather than as failure.

Within the delivered modules, quality is uneven by necessity: authorization and visit longevity received heavy attention; advanced analytics did not. Assessors evaluating the work should weigh invariant correctness above superficial feature breadth.

Repository paths of interest include `Services/`, `Controllers/`, `Data/Migrations/`, `tests/backend/`, and `docs/05-engineering/`. These locations contain the primary evidence for implementation claims made in Chapters 5–7.

Closing of Volume II

Volume II ends here. Together with Volume I (Chapters 1–10, references, appendices) and the supplement on extended engineering detail, the report package aims to present Open Clinical Record as an internship-scale EMR MVP with transparent evidence boundaries. Replace screenshot placeholders, attach signatures, and synchronise any university-specific forms before final binding.

VOLUME III — REQUIREMENTS WORKBOOK, TEST ANNEX, AND DESIGN RATIONALES

Full Functional Requirements Workbook (MVP)

ID	Requirement summary	Trace
FR-001	Register patient with required demographics	See Ch.9
FR-002	Assign unique patient identifier	See Ch.9
FR-003	Search patients by key attributes	See Ch.9
FR-004	View patient profile	See Ch.9
FR-005	Update permitted contact/demographic fields	See Ch.9
FR-006	Flag or manage inactive status	See Ch.9
FR-007	Support deceased recording path	See Ch.9
FR-008	Open patient chart	See Ch.9
FR-009	View allergies list	See Ch.9
FR-010	Add allergy records	See Ch.9
FR-011	View medical history items	See Ch.9
FR-012	Add history items	See Ch.9
FR-013	View visit history list	See Ch.9
FR-014	Open specific visit	See Ch.9
FR-015	Record vital signs for a visit	See Ch.9
FR-016	Record diagnosis entries	See Ch.9
FR-017	Record clinical notes	See Ch.9
FR-018	Finalize a draft visit	See Ch.9
FR-019	Create appointment for patient	See Ch.9
FR-020	View appointment list	See Ch.9
FR-021	Cancel appointment with reason/event	See Ch.9
FR-022	Reschedule appointment	See Ch.9
FR-023	Check in appointment	See Ch.9
FR-024	View appointment events	See Ch.9
FR-025	Authenticate with email/password	See Ch.9
FR-026	Receive JWT	See Ch.9
FR-027	Access /me claims	See Ch.9
FR-028	Enforce role on clinical write	See Ch.9
FR-029	Enforce role on admin routes	See Ch.9
FR-030	Manage users as Admin	See Ch.9
FR-031	View audit events as Admin	See Ch.9
FR-032	View dashboard aggregates	See Ch.9
FR-033	Health liveness	See Ch.9
FR-034	Health readiness with DB check	See Ch.9

Table V3.1 Expanded FR workbook (summaries)

Business Rules Workbook

ID	Rule	Implication
BR-01	Patient registered once	Returns reuse same Patient row
BR-02	Visit non-overwrite	New attendance inserts new ClinicalVisit
BR-03	Check-in creates Draft	CheckedIn creates Draft visit + AppointmentId
BR-04	Final closes visit	Final status ends documentation path for MVP
BR-05	Events on appointment change	Cancel/reschedule/status create AppointmentEvent
BR-06	Server authorization	UI hide is not permission
BR-07	Synthetic demo data	No real PHI in internship artefacts
BR-08	Admin not default clinician	User admin separate from clinical authorship
BR-09	JWT key minimum length	Fail startup if key shorter than 32 chars
BR-10	Development seed gated	Demo seed not implied in Production

Table V3.2 Business rules workbook

NFR Workbook

ID	Theme	MVP interpretation
NFR-S1	Confidentiality	AuthN required for API data access
NFR-S2	Integrity of history	Visits not silently overwritten
NFR-S3	Secret handling	Env-based secrets; no commit of .env
NFR-R1	Controlled transitions	Status changes via services
NFR-M1	Modularity	Services and controllers separation
NFR-M2	Schema evolution	EF migrations under git
NFR-T1	Automated regression	xUnit suite on CI
NFR-T2	Manual PG verification	E2E against PostgreSQL
NFR-U1	Role-aware UX	Navigation reflects responsibilities
NFR-P1	Bounded queries	Take/projection on hot paths

Table V3.3 NFR workbook

Test Case Catalogue (Logical IDs)

ID	Title	Type
TC-001	Health returns ok	Auto/Manual
TC-002	Ready fails when DB down	Auto/Manual
TC-003	Login doctor succeeds	Auto/Manual
TC-004	Login invalid fails	Auto/Manual
TC-005	Receptionist register patient	Auto/Manual
TC-006	Search returns seeded patient	Auto/Manual
TC-007	Unauthorised role rejected on admin API	Auto/Manual
TC-008	Doctor can post clinical documentation	Auto/Manual
TC-009	Nurse can post vitals path	Auto/Manual
TC-010	Create appointment event written	Auto/Manual
TC-011	Cancel writes event	Auto/Manual
TC-012	Reschedule writes event	Auto/Manual
TC-013	Check-in creates draft visit	Auto/Manual
TC-014	Second visit retains first	Auto/Manual
TC-015	Finalize sets Final	Auto/Manual
TC-016	Finalize completes appointment when linked	Auto/Manual
TC-017	Deceased mark role constraints	Auto/Manual
TC-018	Audit entry on sensitive action	Auto/Manual
TC-019	Dashboard loads for authorised role	Auto/Manual
TC-020	JWT missing rejected	Auto/Manual
TC-021	Migration applies on empty DB (manual)	Auto/Manual
TC-022	Seed users exist after Development start	Auto/Manual

Table V3.4 Logical test catalogue

Why REST not GraphQL

REST matched team skills and internship time. GraphQL would add overhead without clear MVP benefit.

Why JWT

SPA-friendly bearer tokens simplify local cross-origin development; cookie hardening remains a deployment topic.

Why four roles

Roles map to clinic jobs; finer permission objects can come later.

Why PostgreSQL

Project architecture decision; strong relational fit for master data and visits.

Why InMemory tests

Speed and isolation for CI; complemented by manual PostgreSQL tests.

Why service layer

Controllers accumulated workflow; extraction clarified transactions and reuse.

Why visit container

Without visits, notes attach ambiguously and history collapses.

Why audit mid-scope

Security-relevant actions needed queryable history for admin trust.

Threat Model Sketch (Lightweight)

A full STRIDE model was not completed. Lightweight threats considered include stolen JWT, privilege escalation via missing Authorize attributes, injection (mitigated by EF parameterisation), secret leakage in git, and over-fetch of clinical data. Penetration testing remains FUTURE WORK.

Operational Runbook Outline

- Provision PostgreSQL 16 and set password environment variable.
- Set OCR_JWT_KEY to a strong value of at least 32 characters.
- Apply migrations (Development auto-migrate or dotnet ef database update).
- Start API; confirm /api/health/ready.
- Start frontend with VITE_API_BASE_URL pointing to the API.
- Use seeded users only in non-production environments.

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Final Annex — Statement on Evidence Integrity

This report package uses placeholders for screenshots and labels [EVIDENCE TO BE INSERTED] where artefacts must come from the student's local environment. Inventing test counts or clinical outcomes would violate academic integrity and professional ethics. Primary evidence remains the GitHub repository and the docs/ tree produced during the internship.

VOLUME IV — CLINICAL WORKFLOW NARRATIVES, UI SPEC NOTES, AND ASSESSOR GUIDE

Reception morning registration

A receptionist authenticates, searches for an existing patient, registers a new synthetic patient when needed, and confirms the MRN is visible on the profile. This establishes identity before any appointment.

Screenshot placeholders in Chapter 6 should be filled with captures of this narrative using synthetic data only.

Scheduling a follow-up

Using an existing patient, the receptionist creates an appointment with date/time context appropriate to the UI, verifies it appears in the list, and confirms an appointment event exists for creation.

Screenshot placeholders in Chapter 6 should be filled with captures of this narrative using synthetic data only.

Patient arrival and check-in

On arrival, check-in transitions the appointment and creates a Draft clinical visit. The chart history should show the new visit without removing prior visits.

Screenshot placeholders in Chapter 6 should be filled with captures of this narrative using synthetic data only.

Nursing vitals capture

A nurse opens the draft visit path, records vitals, and saves. Authorization must allow clinical staff writes while still rejecting unauthenticated calls.

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Doctor consultation

A doctor reviews allergies and history, documents diagnosis and notes, and finalizes the visit. The appointment may move to Completed when linked finalization rules apply.

Screenshot placeholders in Chapter 6 should be filled with captures of this narrative using synthetic data only.

Second attendance proof

The same patient returns on another day. A new visit appears. The first visit remains readable. This is the longitudinal proof point of the MVP.

Screenshot placeholders in Chapter 6 should be filled with captures of this narrative using synthetic data only.

Cancellation path

An appointment is cancelled with validation. Status and event history reflect the change. The patient record itself remains intact.

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Admin oversight

An administrator manages users and inspects audit entries without becoming the default author of clinical notes.

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UI Specification Notes (As Implemented Orientation)

The SPA shell provides role-aware navigation. Login leads to a dashboard appropriate to the role. Patient routes support list/search and detail. Appointment routes support create and operational lists. Clinical routes support chart and visit documentation. Admin routes are restricted. Exact CSS and component names evolve; the architectural intent is stable: one product shell, many role workspaces.

Assessor Reading Guide

Assessors short on time should read: Abstract; Chapters 1, 4, 5, 7, 9, 10; Table 9.1 results matrix; and Appendix limitations. Then sample the repository Controllers, Services, and Tests folders. Screenshots should be evaluated for synthetic data hygiene and for the two-visit longitudinal figure.

Mapping Report Claims to Repository Paths

Claim theme	Primary path
JWT and roles	Extensions/ServiceCollectionExtensions.cs, AppRoles.cs
Migrations	Data/Migrations/
Visit model	Models/Entities/ClinicalVisit.cs, clinical-visit-model.md
Appointment workflow	Services/Appointments/AppointmentWorkflowService.cs
Clinical documentation	Services/Clinical/ClinicalChartService*.cs
Seed/migrate	Data/IdentityDataSeeder.cs
Health	Controllers/HealthController.cs
Automated tests	tests/backend/OpenClinicalRecord.Api.Tests/
CI	/.github/workflows/ci.yml
Frontend API base	src/frontend/.../services/api.ts

Table V4.1 Claim-to-path map

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Q: Does automated testing prove production readiness? A: No. InMemory tests plus manual PostgreSQL checks support regression confidence for covered behaviours only.

Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 23

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Assessor FAQ / Narrative Unit 24

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Assessor FAQ / Narrative Unit 25

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Assessor FAQ / Narrative Unit 26

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

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Assessor FAQ / Narrative Unit 27

Unit 27. Q: Is OCR a hospital-wide EMR? A: No. It is a focused outpatient MVP covering patient management, appointments, and chart/visit documentation with authentication and RBAC.

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

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Assessor FAQ / Narrative Unit 28

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Assessor FAQ / Narrative Unit 29

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Assessor FAQ / Narrative Unit 30

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Assessor FAQ / Narrative Unit 31

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Assessor FAQ / Narrative Unit 32

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 33

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 34

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 35

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 36

Unit 36. Q: Is OCR a hospital-wide EMR? A: No. It is a focused outpatient MVP covering patient management, appointments, and chart/visit documentation with authentication and RBAC.

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 37

Unit 37. Q: Is OCR a hospital-wide EMR? A: No. It is a focused outpatient MVP covering patient management, appointments, and chart/visit documentation with authentication and RBAC.

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Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

Assessor FAQ / Narrative Unit 38

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Assessor FAQ / Narrative Unit 39

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Q: What is the single most important domain rule? A: Patient history is longitudinal; visits are not overwritten when the patient returns.

Q: Where is authorization enforced? A: On the backend via JWT and role policies; the frontend only shapes navigation and controls.

End Matter — Document Control

Field	Value
Document title	Open Clinical Record Industrial Internship Report
Version	1.0 (submission candidate)
Internship window	01–30 September 2026
Host	Infinity Advanced Technology Solutions PLC
Programme	Software Engineering, HiLCoE
Classification	Academic / Industrial Practice
PHI content	None — synthetic only

Table V4.2 Document control

— End of Volume IV — Combined package constitutes the Industrial Internship Report for OCR. —

VOLUME V — DETAILED WEEKLY LOG, DEFECT NOTES, AND SUBMISSION PACK

Indicative Day Log — September 01, 2026

Indicative log entry for day 1 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 02, 2026

Indicative log entry for day 2 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 03, 2026

Indicative log entry for day 3 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 04, 2026

Indicative log entry for day 4 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 05, 2026

Indicative log entry for day 5 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 06, 2026

Indicative log entry for day 6 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 07, 2026

Indicative log entry for day 7 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 08, 2026

Indicative log entry for day 8 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 09, 2026

Indicative log entry for day 9 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 10, 2026

Indicative log entry for day 10 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 11, 2026

Indicative log entry for day 11 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 12, 2026

Indicative log entry for day 12 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 13, 2026

Indicative log entry for day 13 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 14, 2026

Indicative log entry for day 14 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 15, 2026

Indicative log entry for day 15 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 16, 2026

Indicative log entry for day 16 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 17, 2026

Indicative log entry for day 17 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 18, 2026

Indicative log entry for day 18 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 19, 2026

Indicative log entry for day 19 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 20, 2026

Indicative log entry for day 20 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 21, 2026

Indicative log entry for day 21 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 22, 2026

Indicative log entry for day 22 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 23, 2026

Indicative log entry for day 23 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 24, 2026

Indicative log entry for day 24 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 25, 2026

Indicative log entry for day 25 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 26, 2026

Indicative log entry for day 26 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 27, 2026

Indicative log entry for day 27 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 28, 2026

Indicative log entry for day 28 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 29, 2026

Indicative log entry for day 29 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Indicative Day Log — September 30, 2026

Indicative log entry for day 30 of the internship window. Actual daily tasks varied with mentor priorities and defect discovery. Activities across the month included domain reading, requirements refinement, coding, testing, documentation, and demonstration preparation. This indicative log supports the timeline narrative; formal attendance evidence is [EVIDENCE TO BE INSERTED] if required by HiLCoE.

Engineering focus areas recurring through the month: identity and roles; patient APIs; appointment workflow; visit longevity; service extraction; audit; automated tests; CI stability.

Defect and Fix Register (Representative)

ID	Issue	Resolution direction	Status
D-01	Consultation save on closed draft	Harden DocumentVisit path	Addressed in late commits
D-02	EF InMemory dual-attach on visit load	Load by VisitId without broad includes	Addressed
D-03	White page from missing import	Fix AppointmentCreatePage import	Addressed
D-04	Dashboard expensive graphs	Projections / bounds	Ongoing improvement
D-05	Concurrent slot race	Documented limitation	Open / FUTURE

Table V5.1 Representative defects — not an exhaustive production tracker

Submission Pack Contents Checklist

- This PDF report (all volumes bound)
- Signed declaration and approval pages
- Repository tag or commit SHA referenced in submission letter
- Screenshot set replacing placeholders (synthetic data)
- Optional: exported SRS / user manual PDFs already produced in project
- Optional: CI run link
- University industrial practice forms [EVIDENCE TO BE INSERTED]

Concluding Certification of Scope Honesty

The author certifies that this report distinguishes implemented functionality from deferred and out-of-scope items to the best of knowledge based on the Open Clinical Record repository state examined during report preparation. Where evidence could not be verified in-artefact, the label [EVIDENCE TO BE INSERTED] was used rather than fabrication.

— End of Volume V — End of Industrial Internship Report Package for Open Clinical Record —

VOLUME VI — SCREENSHOT REGISTER, FIGURE INDEX, AND FINAL QA CHECKLIST

Screenshot Register (Mandatory Replacements)

ID	Subject	Proves
FIG-6.1b	Repository / IDE tree	Structure proof
FIG-6.2	Patient registration	Create path
FIG-6.3	Patient search	Find path
FIG-6.4	Patient profile	Identity display
FIG-6.5	Appointment create	Schedule path
FIG-6.6	Appointment list	Operational view
FIG-6.7	Check-in	Arrival transition
FIG-6.8	Draft visit	Clinical container
FIG-6.9	Vitals	Observations
FIG-6.10	Notes/diagnosis	Documentation
FIG-6.11	Two visits on chart	Longitudinal proof
FIG-6.12	Finalized visit	Closure
FIG-6.13	Role-specific UI	UX differentiation
FIG-6.14	Audit history	Admin evidence
FIG-6.15	Dashboard	Operational summary
FIG-6.19	Cancellation	Controlled cancel
FIG-6.20	Reschedule/events	History
FIG-7.2	dotnet test output	Automated evidence
FIG-7.3	Authz test evidence	Security tests
FIG-7.4	GitHub Actions	CI evidence

Table V6.1 Screenshot register — replace all placeholders before binding

FIG-6.1b: Repository / IDE tree

Purpose: Structure proof. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.2: Patient registration

Purpose: Create path. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.3: Patient search

Purpose: Find path. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.4: Patient profile

Purpose: Identity display. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.5: Appointment create

Purpose: Schedule path. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.6: Appointment list

Purpose: Operational view. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.7: Check-in

Purpose: Arrival transition. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.8: Draft visit

Purpose: Clinical container. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.9: Vitals

Purpose: Observations. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.10: Notes/diagnosis

Purpose: Documentation. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.11: Two visits on chart

Purpose: Longitudinal proof. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.12: Finalized visit

Purpose: Closure. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.13: Role-specific UI

Purpose: UX differentiation. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.14: Audit history

Purpose: Admin evidence. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.15: Dashboard

Purpose: Operational summary. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.19: Cancellation

Purpose: Controlled cancel. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-6.20: Reschedule/events

Purpose: History. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-7.2: dotnet test output

Purpose: Automated evidence. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-7.3: Authz test evidence

Purpose: Security tests. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

FIG-7.4: GitHub Actions

Purpose: CI evidence. Capture conditions: synthetic data only; no secrets; no real PHI; no tokens visible in browser devtools overlays.

Recommended capture: full application chrome with role indicator visible. Annotate only after export if university requires callouts; keep originals clean.

Final QA Checklist Before University Submission

- Page numbers continuous across bound volumes
- No invented metrics or screenshots
- Role permissions consistent across chapters
- Workflow states consistent (Draft/Final; appointment statuses)
- All OUT OF SCOPE items remain labelled as such
- References cited correspond to real sources
- Appendix references match body mentions
- Student name and signatures applied on declaration/approval
- Repository URL correct
- Synthetic data ethics statement present

Supplementary Index Note 1

Index note 1. Cross-reference reminder: clinical visit longevity is discussed in Chapters 4–6 and tested in LongitudinalVisitTests. Authorization is discussed in Chapters 4–5 and 7 and tested in RoleAuthorizationTests. Results statuses appear in Chapter 9. Recommendations appear in Chapter 10.

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Supplementary Index Note 2

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Supplementary Index Note 3

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Supplementary Index Note 6

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Supplementary Index Note 7

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Supplementary Index Note 8

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Supplementary Index Note 9

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Supplementary Index Note 10

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Supplementary Index Note 11

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Supplementary Index Note 12

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Supplementary Index Note 13

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Supplementary Index Note 14

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Supplementary Index Note 15

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Supplementary Index Note 16

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Supplementary Index Note 17

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Supplementary Index Note 18

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Supplementary Index Note 19

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— End of Volume VI — Report package complete for OCR Industrial Internship —

VOLUME VII — ACKNOWLEDGEMENT OF LIMITATIONS AND FINAL SIGN-OFF NOTES

Closing Reflection 1

Closing reflection 1. The Open Clinical Record internship demonstrated that a software engineering student can, within one month, deliver a coherent outpatient EMR MVP when scope is controlled and domain rules are treated as first-class engineering constraints. The work is incomplete relative to a commercial hospital system—and that incompleteness is documented, not concealed.

Recommendations in Chapter 10 should guide any continuation: concurrency, PostgreSQL CI, frontend tests, deployment hardening, monitoring, document versioning, security assessment, usability evaluation, and continued traceability.

Closing Reflection 2

Closing reflection 2. The Open Clinical Record internship demonstrated that a software engineering student can, within one month, deliver a coherent outpatient EMR MVP when scope is controlled and domain rules are treated as first-class engineering constraints. The work is incomplete relative to a commercial hospital system—and that incompleteness is documented, not concealed.

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Closing Reflection 3

Closing reflection 3. The Open Clinical Record internship demonstrated that a software engineering student can, within one month, deliver a coherent outpatient EMR MVP when scope is controlled and domain rules are treated as first-class engineering constraints. The work is incomplete relative to a commercial hospital system—and that incompleteness is documented, not concealed.

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Closing Reflection 4

Closing reflection 4. The Open Clinical Record internship demonstrated that a software engineering student can, within one month, deliver a coherent outpatient EMR MVP when scope is controlled and domain rules are treated as first-class engineering constraints. The work is incomplete relative to a commercial hospital system—and that incompleteness is documented, not concealed.

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Closing Reflection 5

Closing reflection 5. The Open Clinical Record internship demonstrated that a software engineering student can, within one month, deliver a coherent outpatient EMR MVP when scope is controlled and domain rules are treated as first-class engineering constraints. The work is incomplete relative to a commercial hospital system—and that incompleteness is documented, not concealed.

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Closing Reflection 6

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Closing Reflection 7

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Closing Reflection 8

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Closing Reflection 9

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Closing Reflection 10

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Closing Reflection 11

Closing reflection 11. The Open Clinical Record internship demonstrated that a software engineering student can, within one month, deliver a coherent outpatient EMR MVP when scope is controlled and domain rules are treated as first-class engineering constraints. The work is incomplete relative to a commercial hospital system—and that incompleteness is documented, not concealed.

Recommendations in Chapter 10 should guide any continuation: concurrency, PostgreSQL CI, frontend tests, deployment hardening, monitoring, document versioning, security assessment, usability evaluation, and continued traceability.

Document complete. Student to sign declaration; supervisors to complete approval page; screenshots to be inserted.